A CRITICAL ANALYSIS OF CRYPTOASSETS VALUATION METHOD

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1.0 Introduction

A cryptoasset is most usefully understood as a digital asset whose security, issuance and record of ownership rest upon cryptographic techniques rather than upon the guarantee of any central institution. The category embraces virtual currencies such as Bitcoin alongside a considerably broader family of instruments built upon blockchain technology, a distributed database maintaining an ever-lengthening sequence of ordered records, or blocks, each of which carries a timestamp together with a cryptographic link to its predecessor. It is precisely this architecture that permits blockchain systems to be applied well beyond payments: to the tracking of asset ownership, to the construction of transparent and tamper-resistant voting mechanisms, and to the deployment of decentralised applications (dApps). Cryptoassets are, in the main, designed to function as media of exchange, employing cryptography to secure financial transactions, to verify transfers and to govern the creation of new units; and because no central authority, whether government or bank, stands behind them, they remain, at least in principle, insulated from manipulation and censorship alike.

The population of cryptoassets is far from homogeneous. Virtual currencies of the kind exemplified by Bitcoin and Ethereum serve simultaneously as media of exchange and stores of value, are tied to no single blockchain platform, and circulate across a variety of networks. Security tokens, by contrast, represent claims upon an underlying asset, such as equity, debt or real property, and are engineered to behave in the manner of conventional financial instruments, typically conferring upon their holders an entitlement to financial returns such as dividends or interest (Koffman, 2018). Utility tokens grant access to a product or service, most commonly within the decentralised application upon whose platform they were issued (Yermack, 2017; Howell et al., 2018). To these may be added stablecoins, whose defining ambition is the maintenance of a stable value through a peg to fiat currency or some other asset; non-fungible tokens (NFTs), unique digital assets capable of representing artwork, collectibles and in-game items; and central bank digital currencies (CBDCs), which are digital renderings of fiat money issued and backed by the central bank itself.

The channels through which these instruments come into existence differ accordingly. The major cryptocurrencies are created through mining, whereas utility and security tokens reach their holders through initial coin offerings (ICOs) and security token offerings (STOs) respectively (Liu and Wang, 2019b). Cryptoassets may equally be classified by reference to the blockchain or other technology upon which they are built, their intended use case, and a range of further characteristics.

The present study confines itself to the valuation of cryptocurrency coins and utility tokens. Security tokens are deliberately excluded, for the reason that their economic substance so closely resembles that of traditional financial assets that the valuation apparatus of equity and debt may be applied to them with little modification. The existing literature on cryptocurrencies has concentrated principally upon their role as a payment and transaction mechanism, deriving value from the economics of cryptocurrencies, the properties of the underlying blockchain, and the costs and incentives attaching to mining, with Bitcoin receiving by far the greatest share of scholarly attention (Pagnotta and Buraschi, 2018).

In reviewing and appraising the extant methods for valuing cryptoassets, this report addresses itself to three questions. First, what are the known methods by which cryptoassets may be valued? The answer requires an examination of the several approaches employed to calculate value and of the factors that drive the resulting valuations. Secondly, what form of regression analysis is best suited to the valuation of a cryptoasset? Here the various families of regression are considered with a view to establishing how each may be employed to estimate value. Thirdly, what correlation obtains between Bitcoin, other cryptoassets and traditional assets? The purpose of this final inquiry is a practical one: to establish whether the comparatively rich data available for Bitcoin might be pressed into service in valuing other cryptoassets for which data remain scarce.

2.0 Literature Review

2.1 Fundamentals of Cryptoassets

For the purposes of this research, cryptoassets are organised into three groups, each distinguished by economic function and by regulatory treatment.

The first comprises cryptocurrency coins: digital assets employed as media of exchange within a given cryptocurrency or blockchain ecosystem. Such coins are ordinarily issued by a particular project or platform and are traded on cryptocurrency exchanges against other cryptoassets or against fiat currencies such as the US dollar; Bitcoin and Ethereum furnish the most familiar examples. Coins are distinguishable from tokens in that they commonly serve as means of payment and stores of value, whereas tokens tend to possess a specific use confined to a particular platform. The distinction, however, is not watertight. Although certain coins, Bitcoin above all, circulate as currency well beyond their native environment, many remain bound to the platform of their creation, so that a coin issued by a decentralised application may be usable only within that application and its associated ecosystem. Nor does independence of platform separate the categories cleanly, since both utility and security tokens can exist independently of the platforms upon which they were created and can be traded on exchanges; what ultimately divides the various token classes is the manner of their legal and regulatory treatment rather than any necessary technological affiliation.

The second group consists of utility tokens, which confer upon their holders the right of access to particular products or services. These tokens ordinarily operate within a specific blockchain platform and may be a prerequisite for the use of the services that the platform provides (Catalini and Gans, 2019). A decentralised marketplace, for instance, may require participants to hold and spend its designated token in order to transact at all. Utility tokens thus differ from cryptocurrency coins in that their primary function is not payment or the storage of value but access. It should be observed that while a great many utility tokens have been issued upon Ethereum, whose support for smart contracts has made it the platform of choice for token issuance, this is by no means universal, and other platforms support the creation and circulation

of tokens equally well. The choice of platform, moreover, does not of itself determine a token's usefulness or worth; that depends rather upon the features and capabilities the platform affords.

The third group, security tokens, confer defined financial rights, most typically a share in the profits or revenues generated by a project or asset (Koffman, 2018). Such tokens frequently represent traditional financial assets in tokenised form, whether equities, bonds or real estate, and trade upon cryptocurrency exchanges. Where utility tokens grant access within an ecosystem, security tokens exist to give investors ownership of, and a market in, conventional assets by means of blockchain technology; in consequence they attract a distinct body of law and may require registration with the financial regulatory authorities.

Table 1: Types of cryptoasset (Guesmi and Saadi, 2022)

Cryptocurrency Coins	Utility Tokens	Security Tokens
Payment; smart-contract platform; stablecoin	Payment; access to products or services	Tokenised form of a traditional asset
Mining, token offering	Initial coin offerings (ICOs)	Security token offerings (STOs)

2.2 Tokenomics

Tokenomics denotes the study of the design, behaviour and use of tokens within a given blockchain or cryptocurrency ecosystem. Its subject matter extends from the distribution and allocation of tokens, through their employment as instruments of incentivisation, to the manner of their integration into the wider economy of the platform concerned. The field concerns itself, in essence, with the supply, demand and distribution of tokens and with the incentive mechanisms that drive their use and adoption; it is closely allied to economics proper, yet directs its attention to the peculiar characteristics and dynamics of tokens. An appreciation of tokenomics is indispensable to any assessment of the health and long-run viability of a blockchain project.

The law of supply and demand holds that the price of any good or asset settles at the point where the two forces balance. Applied to tokens, the implication is that value may be driven upward by rising demand, particularly where supply is fixed or otherwise constrained: as more participants

acquire and hold a token, demand presses against a bounded supply and the price tends to appreciate (Liu and Wang, 2019a). The motives underlying such acquisition are various. Some purchasers hold cryptoassets in the expectation of speculative gain; others require them as means of payment or as keys to particular products and services; others still hold them in order to lend support to the underlying project. These motives are illustrative rather than exhaustive, and they differ markedly from one individual to another.

It would nonetheless be a mistake to suppose that supply and demand alone settle the question of value. The health and growth of the underlying platform, the utility of the token itself and the prevailing conditions of the market all exert an influence. Scarcity deserves particular emphasis: where a token is designed to be kept scarce, any appreciation in the value of the network tends to be transmitted to the token, since possession of the token is the condition of access to the network (Klöhn et al., 2018). As the network gains in value, demand for the token may be expected to rise, and its price with it, although the relationship between network value and token value is by no means mechanical, and other factors intervene.

Blockchain projects aspire to assemble a decentralised body of users and participants, a community in the argot of the field. A well-designed token should furnish the means both of incentivising and of monetising the participation of that community, thereby quickening the growth and adoption of the platform and, in turn, the value of the token itself. The incentives on offer take diverse forms: rewards for contributions to the network, privileged access to features or services, or a share in the platform's revenues. By such devices a project may cultivate the engaged and active community upon which its fortunes ultimately depend. Token holders, for their part, are able to dispose of their holdings on secondary markets, namely cryptocurrency exchanges, in exchange for other digital assets such as Bitcoin or Ether, or indeed for fiat money; the existence of such markets renders tokens liquid, allowing holders to realise their value with comparative ease, though the prices obtainable fluctuate and a holder can entertain no assurance of selling at a price to his liking.

Initial coin offerings merit separate remark. An ICO is a species of crowdfunding by which a blockchain project raises capital through the issue of tokens to investors, and among its innovations is the creation of a liquid resale market in those tokens. That liquidity permits holders to monetise the network effects of the project itself, since growth in the platform's

adoption feeds demand for the token and hence its price (Liu and Wang, 2019b). In furnishing token holders with a ready exit, the ICO mechanism has helped to create the vibrant and active markets upon which the growth of many blockchain projects has been built.

2.3 The Cryptoasset Valuation Framework

Notwithstanding a growing body of work, the study of cryptoasset valuation remains at an early stage of development, and a good deal of further research will be required before the determinants of cryptoasset prices are adequately understood. The dynamism of the market itself presents peculiar difficulties for the researcher. Early studies were inclined to treat cryptocurrency markets as creatures of speculation and bubble; yet that characterisation is open to challenge. The absence of direct governmental oversight, and the immunity of these assets from fiscal and monetary policy, do not of themselves entail that prices are driven by speculation alone. The sustained growth of the market, and the widening adoption of cryptocurrencies by individuals and firms, suggest a real and durable demand, although the market's youth counsels caution, and its underlying drivers remain imperfectly mapped.

The sceptical position is most closely associated with Yermack, who identified several weaknesses of Bitcoin considered as a currency and concluded that it possesses no intrinsic value (Yermack, 2015); a number of economists have shared that assessment, finding in Bitcoin's price history the signature of bubble-like behaviour (Cheah and Fry, 2015; Corbet et al., 2018). Yermack's objections were threefold: Bitcoin is not widely accepted as a means of payment; it enjoys the backing of no government or central authority; and its price is subject to violent fluctuation. To these he added the observation that Bitcoin generates no cash flow, and so resists valuation by traditional means. Yet even Yermack conceded that Bitcoin and its peers have attracted remarkable popularity and market value, a concession which implies that forces beyond those captured by the conventional apparatus must be at work. Conley (2017), for his part, argues that acceptance of, and confidence in, the cryptoasset ecosystem is the decisive condition of its capacity to function as a store of value.

Between these poles a spectrum of views has formed. Some analysts maintain that the intrinsic value of Bitcoin is precisely zero, there being no inherent worth and no general acceptance in payment; others have proposed that intrinsic value be located instead in the network effects the

asset commands, in the cost of mining it, or in its engineered scarcity (Pagnotta and Buraschi, 2018). The question of Bitcoin's intrinsic value thus remains contested, and the answer any observer gives is apt to depend upon perspective and investment strategy.

A further strand of the literature contends that the employment of cryptoassets as media of exchange, with the transaction demand and network effects which that use engenders, bears directly upon their value (Peterson, 2017). That work demonstrates that the fundamental factors of usage in trade, money supply and price level each influence the long-term price of Bitcoin, whilst suggesting that the price is likewise driven by investor interest and by the degree of confidence investors repose in the asset.

What emerges from this survey is that the value of a cryptocurrency is jointly determined by considerations both micro and macro in character. At the micro level, the degree of acceptance and confidence within the ecosystem, the use of the asset as a medium of exchange and the merits of the underlying technology all play their part (Vergne, 2017); at the macro level, broader economic conditions, the posture of government regulation and the course of geopolitical events exert their own pressures. Any serious appraisal of a cryptocurrency's value must therefore attend to both registers at once.

Among the micro determinants, the cost of production has attracted sustained attention (Hayes, 2015, 2017, 2019). In the context of cryptocurrencies the cost of production is governed by the computational power required to create new coins, the rate at which they can be produced and the algorithm by which they are generated, the process known as mining. The essential idea is to compute the cost of the electricity and other resources consumed in producing the asset and to treat that cost as a reference point for its value. Cost-of-production valuation is of respectable lineage, having long been employed in the appraisal of goods and commodities (agricultural produce, minerals and industrial products among them) by estimating the labour, materials and capital their production requires. The method is serviceable but imperfect: its accuracy is hostage to movements in input costs over time and to the efficiency of the production process, and it is accordingly best deployed in conjunction with other valuation methods rather than in isolation.

Table 2: Valuation framework for cryptoassets (Guesmi and Saadi, 2022)

Approach	Methods

Monetary approach	The crypto quantity theory of money
Income approach	Quantity theory of money; crypto J-curve methodology; network models and Metcalfe's law
Market approach	Network value-to-transaction ratio (NVT); price-to-utility ratio
Cost approach	Cost of production (mining); market-value-to-realised-value ratio (MVRV); market-value-to-thermo-value ratio; investor capitalisation; short-to-long-term realised value ratio
Profit-loss approach	Realised profits-to-value ratio (RPVR); short-term-holder profit/loss ratio (STHPL); seller exhaustion

3.0 Cryptoasset Valuation: The Income Approach

3.1 The Crypto Quantity Theory of Money (QTM)

The quantity theory of money is a macroeconomic proposition describing the relation between the quantity of money circulating in an economy and the general level of prices for goods and services (Friedman, 1956), and it has long been invoked to explain the behaviour of prices and inflation over time. Its canonical statement is the equation of exchange:

$$MV = PQ$$

in which M denotes the quantity of money in the economy; V the velocity of money, that is, the rate at which money passes from one transaction to the next; P the general price level of goods and services; and Q the quantity of goods and services produced. The equation asserts that the total value of economic transactions, MV, equals the general price level multiplied by the volume of output. In empirical practice real gross domestic product is commonly substituted for Q, since it affords a more accurate and comprehensive measure of the volume of goods and services an economy produces, whereupon the identity becomes:

$$MV = P \times \text{Real Gross Domestic Product (GDP)}$$

Transposed to the setting of cryptocurrency coins and utility tokens, the QTM framework offers a means of appraising value and the prospects of appreciation or depreciation. The theory implies that an increase in the quantity of money, here the number of coins or tokens on issue, produces a corresponding rise in the general price level, and a contraction the reverse. Applied to cryptoassets the equation retains its familiar form,

$$MV = PQ$$

save that M now denotes the quantity of coins or tokens in circulation; V their velocity, the rate at which they change hands; P the general price level of the goods and services purchasable with them; and Q the quantity of such goods and services. The total value of transactions conducted in the token, MV, thus equals the price level of token-denominated goods and services multiplied by their quantity.

A point of interpretation deserves emphasis. Within this framework P is the price of goods and services *expressed in the token*, and is therefore the reciprocal of the token's own price: it records the quantity of goods and services a unit of the token will command. In the QTM equation for cryptoassets, accordingly:

$$\text{Token Price} = \frac{1}{P} = \frac{Q}{MV}$$

By way of illustration, a token trading at \$10 implies $P = 1/10 = 0.1$, signifying that a unit of the token purchases 0.1 units of goods or services. For utility tokens the velocity term should equate to the overall value of the services rendered through the blockchain network during a specified period: where a token is spent upon goods or services within the network, it is the value of those goods and services, Q, that ought to enter the computation of the token's transactional worth. The term Q should therefore be read, for utility tokens, as the aggregate value of network services over the period, and it is from this quantity that the velocity V falls to be calculated.

As to the money stock, M denotes the quantity of coins or tokens in circulation. In many cases the total supply, M^*, is fixed by the issuer at the outset, in which event the total supply is properly treated as a constant for valuation purposes, circulation being some fraction f of it:

$$M = M^* \times f$$

Not every cryptoasset, however, enjoys a fixed total supply: where supply expands over time, whether through mining or otherwise, M ceases to be constant and the equation must be modified to accommodate its evolution. The valuation expression then reads:

$$\text{Token Price} = \frac{1}{P} = \frac{Q}{MV} = \frac{Q}{M^* \times f \times V}$$

On the QTM account, therefore, the value of a token is determined by its utility, its velocity and the quantity on issue; and since the quantity in circulation is fixed in the long run, it is utility and velocity that do the real work. Tokens of higher utility, measured by the overall value of services rendered through the network, tend to command higher values, being in greater demand for transactional purposes; tokens of lower velocity, held for longer periods by investors speculating

upon their future worth, tend likewise to appreciate. The framework accordingly predicts that value accrues most readily to tokens combining high utility with low velocity.

The QTM approach bears a family resemblance to the classic discounted cash flow method of asset valuation, and may fairly be located within the income approach, which rests valuation upon the expected future cash flows of the asset concerned. Discounted cash flow analysis values an asset by projecting its future cash flows and discounting them to present value at a rate reflecting both the time value of money and the riskiness of the asset. A firm appraising a new product, for example, might forecast the product's cash flows and discount them to obtain an estimate of its worth. The method is powerful but not infallible: its output is only as reliable as the forecast cash flows and the assumptions embedded in the discount rate, for which reason it is prudently used alongside other valuation methods rather than in their stead.

Cryptoassets present the DCF practitioner with an obvious embarrassment: they generate neither cash flows nor earnings in the manner of traditional assets, and the standard machinery cannot be applied to them without adaptation. A modified DCF is nonetheless conceivable, in which the analyst estimates the future flows attributable to the cryptoasset and discounts them at a rate appropriate to its risk, thereby deriving an estimate of potential value grounded in expected future flows. The difficulties are not trivial, for the flows of a cryptoasset resist confident forecasting and its value is volatile and at the mercy of market forces; the method should accordingly serve as one element within a larger valuation exercise rather than as a self-sufficient answer.

3.2 Crypto J-Curve Methodology

The J-curve is commonly employed to illustrate the interplay between the current utility value (CUV) of an asset and its discounted expected utility value (DEUV). Consider a hypothetical asset with a CUV of $100 and a DEUV of $200. The J-curve locates the asset's initial value at the intersection of the two, namely $100; thereafter the value may rise or fall as CUV and DEUV themselves evolve. Should the CUV advance to $150 while the DEUV holds steady, the asset's value rises to $150; should the DEUV instead decline to $150 with CUV unchanged, the value falls to the same figure. The instrument thus captures the dynamic relation between present

utility and discounted expectation, and the manner in which the two jointly determine value through time.

The crypto J-curve framework proposes, in a manner analogous to its traditional counterpart, that the value of a cryptoasset is determined by two components (Burniske, 2017). The first is the current utility value, representing that portion of value derived from the utility and usage of the asset in the present. It is governed by such factors as the number of users, the degree of adoption and the usefulness of the asset in performing particular tasks or delivering particular services (Burniske, 2017), and it may be expected to shift over time as the project develops and its usage evolves. Formally:

$$CUV = \frac{U_t}{MV} = \frac{\text{Token's current supply}}{\text{Token supply} \times \text{Token velocity}}$$

The second is the discounted expected utility value, representing the value derived from investment speculation directed at the future. The DEUV embodies investors' expectations of the flows the asset will in due course generate, discounted to present value, and it fluctuates as market perceptions of the project's prospects mature. Burniske's DEUV may be regarded as a modified rendering of the DCF method for cryptoassets: in place of forecast cash flows it captures the value assigned by speculative expectation. It is obtained residually, as the excess of price over current utility value:

$$DEUV = \text{Price} - CUV = \text{Price} - \frac{\text{Token's current supply}}{\text{Token supply} \times \text{Token velocity}}$$

$$DEUV = \text{Price} - \frac{U_t}{MV}$$

The two components take turns, as it were, in governing the token's value as a blockchain project develops and market perceptions of it change. In a project's infancy, value is apt to rest chiefly upon CUV, investors prizing the utility the token presently affords; as the project matures and its potential becomes more legible, the balance shifts towards DEUV, with investors speculating upon the flows the token is expected in future to generate. It is this alternation that the crypto J-curve is designed to capture, and the framework falls squarely within the income approach to cryptoasset valuation, oriented as it is to the future flows the asset is expected to yield.

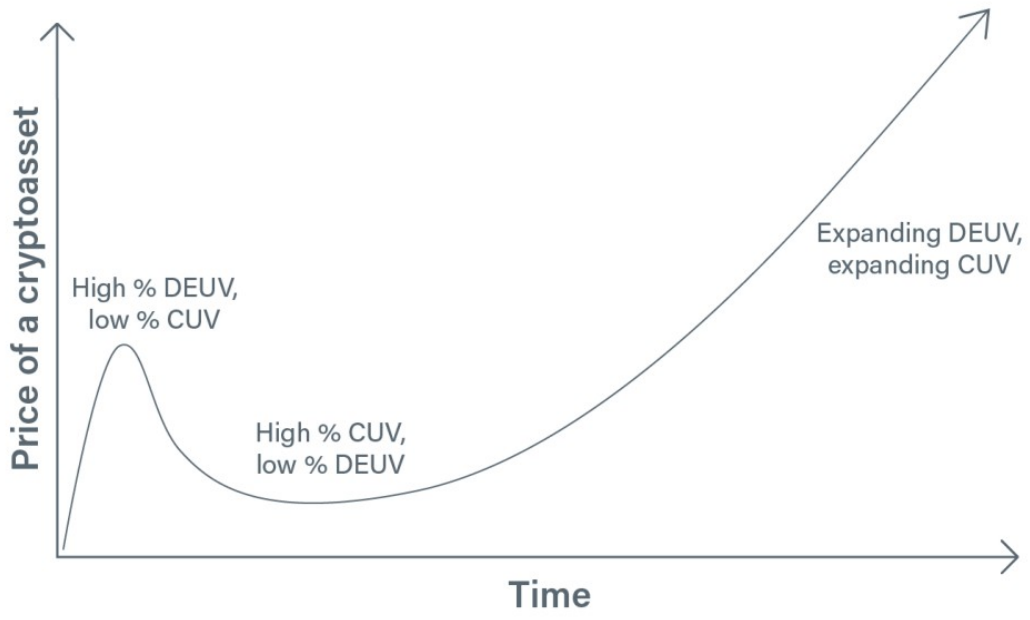


Figure 1: The crypto J-curve (The crypto J-curve)

Certain adopters of the framework have gone further, employing it as a proxy for the successive life stages of a cryptoasset. Early in a project's existence, on this reading, value is driven principally by current utility; with maturity, and as the prospect of future growth clarifies, discounted expectation assumes the greater weight. Tracked over time, the movements of CUV and DEUV thus illuminate the forces bearing upon the asset's value at each stage of its development.

3.3 Network Models and Metcalfe's Law

Metcalfe's law holds that the value of a network is proportional to the square of the number of its users. First advanced by Robert Metcalfe in the 1980s, the principle has since been applied to networked systems of many kinds, including social networks, computer networks and telecommunications systems, and its central implication is that network value rises, disproportionately, as membership grows (Peterson, 2018).

The logic is combinatorial. A network of n users admits $n(n-1)/2$ possible connections, which is the number of distinct pairings its members can form. Metcalfe's law takes the value of the network to be proportional to n^2, so that:

$$\text{Value of Network} = k \times n^2$$

where k is a constant representing the value of each connection, estimable by reference to the value of comparable networks or through market research.

Applied to cryptoassets, the law suggests that the value of an asset stands in relation to the number of users, and the degree of adoption, of the network upon which it is built (Metcalfe, 2013). The framework would seem particularly congenial to Bitcoin, whose general application as a method of payment and whose peer-to-peer architecture, which permits users to transact directly with one another without intermediaries, give its network effects unusual prominence. Peterson's formulation values the Bitcoin network as:

$$\text{Value of Bitcoins}, V = a \times \left(\frac{n(n-1)}{2} \times \frac{1}{b_t} \right)$$

$$\text{The number of Bitcoins over time}, b_t = b_{t-1} \times \ln\left(\frac{1}{b_{t-1}} \right)$$

where a is the beta coefficient of the linear regression model set out below, and B is the total stock of Bitcoin (21 million):

$$\ln\ln(Y_t) = \alpha + \beta \left[\frac{\ln(B/b_t)}{\ln(M_t)} \right] + \varepsilon_t$$

Peterson (2018) is candid about two limitations attending the use of Metcalfe's law as a predictor of Bitcoin's value. The first is one of horizon: the approach speaks only to the medium and long term, and makes no attempt to account for short-run price movements, which answer to a far wider range of influences. The second is one of reliability: the law has not always predicted Bitcoin's value accurately, the period from 2013 to 2014, when prices departed from anything resembling normal supply and demand under fair competition, furnishing the standard counterexample. Metcalfe's law is, after all, a general principle, blind to the specific factors that may bear upon the value of Bitcoin or of any other network; a prudent appraisal will therefore weigh a variety of considerations rather than rest upon the law alone.

4.0 Cryptoasset Valuation: The Market Approach

4.1 Network Value-to-Transaction Ratio (NVT)

The network value-to-transaction ratio measures the value of a network relative to the volume of economic activity conducted upon it, and is computed by dividing the network's market capitalisation, the aggregate value of all units of its currency, by the daily transaction volume passing across it (Woo, 2017):

$$NVT\ Ratio = \frac{\text{Network Value}}{\text{Daily Transaction Volume}}$$

The ratio has become a standard valuation metric for cryptocurrencies precisely because it permits comparison across networks on the footing of the economic activity each supports. A high NVT signifies a market capitalisation large relative to the network's transactional activity, and may be read as a symptom of overvaluation; a low NVT, conversely, suggests a capitalisation modest relative to activity, and hence possible undervaluation. The metric is nonetheless imperfect, and a valuation resting upon it alone would be imprudently narrow.

Its limitations repay careful statement. In the first place, the ratio attends only to market capitalisation and transaction volume, and is silent as to the further considerations that bear upon a network's value: the utility of its currency, the trust and adoption it commands among users, and the general condition of the market (Liu, 2019; CAIA, 2018). In the second place, the NVT rests upon the assumption that a cryptoasset derives its value primarily from service as a medium of exchange; it takes no account of the store-of-value function, and may therefore misprice assets held chiefly as stores of value rather than for transactional use. Thirdly, there is the matter of timing: Woo (2017) himself observes that the ratio is ineffective as an advance detector of price bubbles, since market capitalisation and transaction volume may not faithfully reflect underlying value; its proper office, on his account, is the retrospective one of discriminating between price collapse and price consolidation after the event, rather than the prediction of future movements. Fourthly, the ratio is of limited service in the short run, its usefulness lying chiefly in longer-horizon assessment, for short-term price movements answer to a range of influences the metric does not capture. Finally, comparisons across networks must be drawn with care, since the ratio

is sensitive to the manner in which transaction volume is measured, and different measurement conventions can yield significantly different results.

For all these reasons, while the NVT ratio yields genuine insight into the value of a network, it ought not to be relied upon as the sole determinant of the value of any cryptocurrency, and other factors must enter the appraisal.

4.2 Price-to-Utility Ratio

The price-to-utility ratio (the PUT, or P/U, ratio) measures the value of a cryptoasset relative to the utility of its underlying network, and is obtained by dividing the asset's current market price by the estimated utility of the network as reflected in such factors as the number of users, the volume of transactions and the general level of adoption and usage. Liu (2019) advances the P/U ratio expressly as a remedy for the shortcomings of the NVT ratio, which considers only market capitalisation and transaction volume and neglects the potential of a cryptoasset to serve as a store of value:

$$\text{Price-to-utility Ratio}\left(\frac{P}{U}\right)=\frac{\text{Price Per Token}}{\text{Utility Per Token}}=\frac{\text{Market Cap}}{\text{Token Utility}}$$

By setting market price against network utility, the P/U ratio incorporates both the medium-of-exchange and the store-of-value functions of a cryptoasset, and thereby furnishes the more comprehensive valuation. Its reading follows the familiar pattern: a high ratio indicates a price elevated relative to estimated utility, and hence possible overvaluation; a low ratio, a price depressed relative to utility, and hence possible undervaluation.

The total utility of a token answers to many influences, of which three are paramount: token velocity, the staking ratio and the dilution rate. Token velocity measures the speed at which tokens circulate within an ecosystem; the higher the velocity, the more actively the tokens are employed, which may betoken higher levels of engagement and adoption. Formally:

$$\text{Token Utility}=\frac{\text{Token Velocity}\times\text{Staking Ratio}}{\text{Dilution Rate}}$$

The staking ratio records the proportion of tokens locked up, or staked, rather than actively traded or spent. Staked tokens are withdrawn from use, yet may earn rewards for those who stake

them, and a high staking ratio signals that users are placing long-term faith in the project and are content to commit their tokens for extended periods; such confidence enhances the token's utility as a store of value:

$$\text{Staking Ratio} = \frac{\text{Staked Tokens}}{\text{Total Token Supply}}$$

The dilution rate, finally, is the pace at which new tokens enter the ecosystem, swelling supply and, potentially, depressing the value of each unit outstanding. The Bitcoin halving of 2020 furnishes an instructive example: the dilution rate fell below two per cent, the inflation target of many major central banks (Liu, 2019), a development fairly read as favourable to Bitcoin, since it implies a supply not being inflated with undue haste and hence a value more readily maintained.

Liu (2019), applying the token utility formula to Bitcoin, found that the asset's utility had risen appreciably over the preceding five years, a finding consistent with the proposition that token utility leads token price. The increase he attributes to a rise in velocity and in the staking ratio, accompanied by a decline in the dilution rate; fewer liquid tokens circulated than the market demanded, and the utility of the remainder was bid upward accordingly. He shows, further, that a contraction in the stock of liquid Bitcoin, together with rising velocity, contributed to ending the bear market of 2018; and that Bitcoin's P/U ratio ordinarily inhabits a range of 50 to 200 under normal market conditions, rising above 200 in bubble episodes and falling below 50 in the depths of a crypto winter. These regularities suggest that the P/U ratio can serve as a genuine diagnostic of market condition: by setting price against utility, investors and analysts may judge whether a token stands dear or cheap relative to its potential, to the benefit both of investment decisions and of any rounded picture of the token's state and prospects.

The P/U ratio was conceived to relieve certain difficulties attending the NVT ratio, notably the difficulty of identifying comparable projects, yet it labours under some of the same constraints. Chief among them is heterogeneity: blockchain projects address different problems, deliver different services and court different constituencies of users, so that the utility of one token cannot be set against that of another without strain. A project may combine high velocity and a high staking ratio, the marks on paper of high utility, while serving a niche market of limited growth; another may post more modest readings while addressing a broad market of far greater

potential. In such a comparison the ratio alone will not say which project possesses the higher utility. The P/U ratio is therefore best treated as one instrument among several, to be applied with due regard to the particular circumstances of each project.

5.0 Cryptoasset Valuation: The Cost Approach

5.1 Cost of Production (Mining)

The cost approach values an asset, of which real estate and going concerns are the traditional subjects, by estimating what it would cost to reproduce or replace it, upon the premise that an asset is worth no more than the cost of creating its like from scratch. The appraiser proceeding under this approach first determines the cost of reproduction or replacement, taking account of materials, labour and associated factors, and then adjusts that figure for the age and condition of the asset and for any movement in market conditions or construction costs; the adjusted figure is the asset's value under the cost approach. In practice the method is seldom used alone, being paired with the market and income approaches so as to yield a more complete picture, and it proves especially serviceable for assets that are unique or for which market data are thin.

The cost-of-production framework for valuing cryptocurrencies and other digital assets was developed in 2015 by the economist and blockchain researcher William Mougayar. Its animating idea is that the value of a digital asset should rest upon the cost of creating, or mining, the asset: the electricity and other resources consumed in running the network and verifying transactions, together with the hardware and equipment these operations require. To apply the model, the appraiser estimates the total cost of producing a given asset and divides by the number of units produced, the resulting per-unit cost standing as the asset's value under the model.

Cryptocurrency mining employs specialised computers to verify transactions and append them to the blockchain, an activity whose appetite for electricity and other resources accounts for a substantial share of its cost. The model decomposes that cost into two elements. Fixed costs are those required to establish and maintain the operation (the purchase and installation of mining machines and their transport to the mining site) and remain broadly invariant to output. Variable costs move with production, and in cryptocurrency mining the dominant variable cost is electricity, which is required to run the machines and verify transactions and which is, significantly, paid in national currency such as dollars or euros, rather than in the mined asset itself. Miners must therefore have regard to the fiat price of the asset in judging whether their

operations remain profitable, the more so as electricity costs themselves vary with local prices and with the efficiency of the machinery employed.

The economic logic rests upon the assumption that miners operate in a competitive market and are moved by the expectation of profit. So long as the market price of the mined asset exceeds its cost of production, mining continues, since it pays; should price fall below cost, the incentive weakens and activity may be expected to contract. The market price of a digital asset accordingly plays a decisive role in determining the level of mining activity: high prices and low costs summon forth effort, while low prices and high costs discourage it.

Hayes (2015) offers an estimating equation for the daily cost of production:

$$\text{The Daily Production Cost}, E_{day} = \text{Electricity Cost} \times \text{Mining Hours Per Day} \times \text{Hash Rate} \times \text{Energy Efficiency}$$

Each element merits comment. The electricity cost per kilowatt-hour weighs heavily, mining being energy-intensive, and dearer power translates directly into dearer production. The number of operating hours matters likewise, mining operations typically running twenty-four hours a day, seven days a week, so that longer running times raise the daily cost. The hash rate, the computational effort expended in verifying transactions and appending blocks, raises both the probability of solving a block and earning its reward and, with it, the electricity consumed. The average energy efficiency of the machines, measuring energy consumed per unit of mining effort, completes the calculation, more efficient machinery improving the prospect of reward for a given expenditure.

The expected number of bitcoins produced each day is fixed by the Bitcoin algorithm devised by the currency's creator (Nakamoto, 2008), under which daily issuance halves approximately every four years in the process known as the halving. Daily production has therefore declined steadily from the generous issuance of Bitcoin's early years, and at the time of writing stands at approximately 1,800 coins per day, a figure itself subject to change with the circumstances of the network and the wider market. The expected level of daily production when mining directly is given by:

$$BTC/day = \left(\frac{\text{Current Block Reward} \times \text{Hash Rate} \times \text{Sec}^{-1}}{\text{Mining Difficulty} \times \text{Probability of Winning a Block}} \right) \times \text{Mining Hours Per Day}$$

$$BTC / day = \left(\frac{\text{Current Block Reward} \times \text{Hash Rate} \times 3600}{\text{Mining Difficulty} \times \frac{1}{2^{32}}} \right) \times 24$$

Four determinants govern this expression. The current block reward, the number of bitcoins awarded to the miner who solves a block, stands at present at 6.25 BTC, though the figure falls at each halving. The hash rate per second measures the computational power deployed, higher rates improving the odds of capturing the reward. The mining difficulty registers the complexity of the cryptographic puzzles the machines must solve; as difficulty rises, blocks become harder to win and expected daily production falls, the probability of solving any given block being $1/2^{32}$, or one chance in 4,294,967,296. The number of mining hours per day completes the account, most operations running continuously.

In a competitive market the marginal product of mining should in theory equal its marginal cost, which should in turn equal its selling price; the objective price of production, the price at which the miner can sell the mined asset, should therefore equal the cost of producing it:

$$\text{The Objective Price of Production Level}, p^* = \frac{E_{day}}{Bitcoin / Day}$$

Whether Bitcoin possesses intrinsic value by virtue of its limited supply and usefulness in exchange has been argued in the affirmative; yet its realised value is ultimately set by market demand. Hayes (2017) contends that opportunities for excess returns in the mining industry are short-lived precisely because the industry is competitive: miners perpetually seek out the most profitable coins to mine, driving up aggregate network hashing power and, with it, mining difficulty; as difficulty rises, profitability per unit of effort declines, and the prices of the coins being mined fluctuate as participants dispose of their relatively overvalued holdings. Sustained super-normal profit is therefore elusive. Estimated on this footing, Hayes (2017) reports that the cost-of-production model explains approximately 92 per cent of the observed market price of Bitcoin. Corroboration is available from several quarters: Gmitro (2017), examining the technological and economic incentives behind cryptocurrency development through a market-based analysis, computes an objective price of production of $1,875.65; Li and Wang (2017) find mining cost empirically relevant to cryptoasset valuation; Bhambhwani et al. (2019) identify computing power as a significant determinant of value, high computing power improving

transaction speed and security and so fortifying trust in the blockchain; and Garratt and van Oordt (2019) observe that currencies mined with application-specific integrated circuits (ASICs), Bitcoin among them, prove less vulnerable to adverse exchange-rate shocks than their peers.

The model, for all its explanatory success, carries assumptions that limit its reach. It presumes, first, a perfectly competitive market in which every miner is a price-taker; yet the expense of mining equipment begets economies of scale, and economies of scale beget centralisation, which bears both upon the market price of the currency and upon the individual miner's cost of production (Singh and John, 2019); the textbook market and the actual one are not the same. It presumes, secondly, that miners answer only to profit and will withdraw when marginal cost exceeds price; but some miners are sustained by conviction, by a belief in the principles of an unregulated and transparent financial system, and will mine at a loss to support and develop the network (Ige, 2018), so that the model may mispredict their behaviour. Thirdly, Hayes' construction is native to Bitcoin and to proof-of-work systems resting upon electricity-based mining; it adapts tolerably to other such systems, but sits ill with proof-of-stake arrangements, in which consensus is established by reference to a forger's relative network stake and electricity cost is simply not a relevant input (Singh and John, 2019). The valuation of such assets requires attention to the specific consensus algorithm and to the further factors peculiar to them.

5.2 Market-Value-to-Realised-Value Ratio

The MVRV ratio, a metric employed in the analysis of the Bitcoin market, is calculated by dividing the market capitalisation of Bitcoin by its realised capitalisation, and measures the relation between the current price and the average cost basis of all participants in the market:

$$MVRV\ ratio = \frac{\text{Market Capitalisation}}{\text{Realised Capitalisation}}$$

Allied to it is the MVRV Z-score, which measures the deviation of market capitalisation from realised capitalisation, scaled by the standard deviation of market capitalisation (Bitcoinition; LookIntoBitcoin):

$$MVRV\ Z\text{-}Score = \frac{\text{Market Cap}_{USD} - \text{Realised Cap}_{USD}}{\text{StdDev}\left(\text{Market Cap}_{USD}\right)}$$

The Z-score serves to identify prospective market tops and bottoms by registering how far capitalisation has strayed from the market's average cost basis. A high reading indicates a market capitalisation standing well above realised capitalisation, a possible symptom of overvaluation and impending correction; a low reading indicates the reverse, and may presage a rally. The score is, however, one instrument among many, and should be weighed alongside other factors rather than made the sole basis of an investment decision.

An MVRV ratio below unity indicates a market selling, in aggregate, at a loss, historically the signature of a cyclical bottom. Conversely, when Bitcoin's market capitalisation swells dramatically relative to its average cost basis, conditions are ripe for profit-taking, and holders may find the moment opportune to realise gains (Dodd, 2021). The caveat entered above applies with equal force here: the MVRV ratio has served well in the past, but it is not infallible, and it ought not to be relied upon in isolation.

5.3 Market-Value-to-Thermo-Value Ratio

The market-value-to-thermo-value (MVTV) ratio sets the market value of an asset against its thermo-value. In its general form the thermo-value of an asset measures the maximum energy extractable from it through such processes as combustion or decomposition, and the ratio serves as a gauge of sustainability: a high MVTV indicates a market value far in excess of thermo-value, and hence a market possibly overvaluing the asset and exposed to correction, while a low MVTV suggests undervaluation and the prospect of a rally (Dodd, 2021).

In its application to Bitcoin, the MVTV is a mean-reversion metric calculated by dividing the market capitalisation of Bitcoin by its thermo-capitalisation, the cumulative dollar value paid to miners for securing the network. The ratio thus compares the price of Bitcoin with the implied value paid to the miners who validate transactions and secure the chain, and it may be thought of as the crypto analogue of the EV-to-EBITDA multiple of equity valuation, setting investors' current market capitalisation against miners' cash flow. Its diagnostic use follows the established pattern: a high reading marks a market value standing well above the network's thermo-value, and hence possible overvaluation; a low reading, possible undervaluation and the conditions for recovery.

5.4 Investor Capitalisation

Investor capitalisation measures the value of Bitcoin held by investors, and is obtained by subtracting thermo-capitalisation from realised capitalisation. By stripping from the aggregate cost basis the outstanding value paid to miners, the metric isolates the value attributable to investors proper, and in doing so furnishes a gauge of capitulation during bear markets, permitting an assessment of Bitcoin's fair value at the bottom of a cycle. A depressed investor capitalisation may signal an undervalued market ripe for recovery (Dodd, 2021).

The behaviour of the metric across the cycle is itself informative. In bear markets the total value of all bitcoins in circulation, the market capitalisation, tends to converge upon investor capitalisation, and the two measures have typically inflected when near parity. In October 2021 investor capitalisation stood at approximately $400 billion, some 68 per cent below market capitalisation, a gap suggesting a market rich relative to the value of coins in investors' hands, and correspondingly exposed to correction. As ever, such metrics are most safely read alongside other evidence rather than in isolation.

5.5 Short-to-Long-Term Realised Value Ratio

The SLRV ratio addresses the behaviour of Bitcoin holders. It is calculated by dividing the one-day HODL wave by the six-month-to-one-year HODL wave, both weighted by realised capitalisation. The HODL wave measures the holding periods of Bitcoin owners: the one-day wave records the percentage of coins held for at least one day, the six-month-to-one-year wave the percentage held for between six months and a year, and the weighting by realised capitalisation renders the resulting ratio a more faithful representation of holder behaviour.

Read as a market signal, a high SLRV indicates that a large share of coins is being held for extended periods, a condition that has accompanied overvaluation and impending correction, while a low reading marks the opposite condition and has preceded rallies. The ratio also serves to expose potential price inefficiencies: historically, readings below 0.04 have coincided with bear markets, in which short-term velocity is low and investors' appetite for trading has ebbed, circumstances suggesting an oversold market with room to recover.

6.0 Cryptoasset Valuation: Profit and Loss Metrics

6.1 Realised Profits-to-Value Ratio

The realised profits-to-value (RPV) ratio sets the realised profits of Bitcoin against its market value, realised profits being computed as the excess of current market value over the cost basis of the asset. As a gauge of the market's sustainability the ratio reads in the customary manner: a high RPV, marking realised profits large relative to market value, hints at overvaluation and correction to come; a low RPV, at undervaluation and the prospect of a rally.

The boundary cases illuminate the metric. A ratio of unity would signify that every bitcoin in existence had been traded within a single day, an event without precedent in the history of the market, and a practical impossibility given the fixed supply, since the number of coins tradable in a day is bounded by the market volume available at any moment. The historically informative thresholds lie elsewhere. When profit-takers have moved two per cent or more of their holdings within a single day, the market has tended to stand at a cyclical top: profit-taking on that scale betrays a rising risk-aversion among investors, whose disposals depress demand and, with it, price. When, by contrast, profit-takers have moved a mere 0.001 per cent of their coins in a day, the market has typically been in the process of carving out a bottom: reticence on that scale suggests investors disposed to accumulate rather than to sell, and demand so generated tends to carry price upward.

6.2 Short-Term-Holder Profit/Loss Ratio

The short-term-holder profit/loss (STHPL) ratio likewise concerns the behaviour of Bitcoin holders, comparing the fortunes of short-term holders, those in possession of their coins for less than one year, with those of long-term holders, whose coins have been held for longer. The ratio is obtained by dividing the profit or loss of the former by that of the latter.

Its diagnostic value lies in the identification of market tops and bottoms. A high reading indicates short-term holders faring conspicuously better than their long-term counterparts, a condition associated with overvaluation and vulnerability to correction, while a low reading marks the reverse and has tended to accompany undervalued markets on the eve of recovery. A ratio of

unity typically marks a local bottom in a bull market or a local top in a bear market. Below unity, the aggregate position of participants who have moved coins within the preceding 155 days is loss-making, a possible sign of an oversold market due a rally; above unity, short-term participants stand in aggregate gain, and the market may be judged overbought and exposed to correction.

6.3 Seller Exhaustion

Seller exhaustion denotes the condition in which the supply of an asset offered for sale is reduced or depleted, whether because sellers have exhausted their holdings or because they have lost the inclination to sell. In such a market demand may persist undiminished while the ranks of willing sellers thin, and price rises as buyers bid for what remains of the available supply.

In the Bitcoin context the concept is usefully paired with volatility, understood as the degree to which price fluctuates over time: a highly volatile market is one in which price can swing substantially within a short interval, a less volatile one that in which movement is slower and steadier. Taken together with profit metrics, volatility offers a serviceable guide to Bitcoin’s future price action. Elevated volatility bespeaks an uncertain and unpredictable market in which the direction of price resists forecasting; subdued volatility bespeaks a steadier market whose price may be expected to continue upon its established course.

7.0 Regression Analysis

7.1 Methodology

Linear regression admits of several varieties, of which simple and multiple regression are the principal. Simple linear regression models the relation between two variables where a linear association subsists between them, a single independent variable being employed to predict the value of the dependent variable. Multiple regression extends the apparatus to two or more independent variables, furnishing a statistical method by which the joint effects of several predictors upon a single response variable may be examined; a separate regression coefficient is estimated for each independent variable, registering the strength and direction of its relationship with the dependent variable, and the fitted coefficients may then be used to predict the dependent variable from known values of the predictors.

The present research undertakes a time-series analysis and prediction of the closing price of Bitcoin by means of three regression models implemented in the Python programming language: linear regression, support vector regression (SVR) and a neural network. The Python implementation first loads the requisite libraries and reads a CSV file containing the historical closing prices of Bitcoin; the data are sorted by date and plotted; training data are then constructed for the three models, each of which is trained upon them; and each trained model is finally employed to generate predictions of future closing prices, which are plotted against the actual series for inspection.

7.1.1 Linear regression

Linear regression is a statistical method for modelling the relation between a dependent variable and one or more independent variables, its name deriving from the assumption that the relation between the variables is linear, namely that the dependent variable is a linear function of the independents. The fitted line may be written:

$$\hat{y} = a + bx$$

where y is the dependent variable, x the independent variable, a the intercept and b the slope of the line, the least-squares estimates of which are:

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

The object of the exercise is the line of best fit from which predictions of the dependent variable may be made given values of the independents. That line is found by minimising the sum of squared differences between the observed values of the dependent variable and those the model predicts (the method of least squares), the solution being obtained from the system known as the normal equations. The technique is a staple of statistical analysis wherever the relation between a dependent variable and one or more predictors is at issue.

The regression coefficients, the values multiplying the independent variables in the fitted equation, record the strength and direction of each predictor's relationship with the response. A positive coefficient signifies that the dependent variable tends to rise with the predictor; a negative coefficient, that it tends to fall as the predictor rises; and the magnitude of the coefficient registers the strength of the association, a larger coefficient implying that a small movement in the predictor produces a larger movement in the response (T.S., 2020).

Certain summary statistics attend the fitted model. The coefficient of determination, R^2, measures the proportion of variance in the dependent variable explained by the predictors; computed as the square of the correlation between observed and predicted values, it ranges from zero, for a model explaining none of the variance, to unity, for a model explaining it all, higher values indicating the better fit. The standard error measures the uncertainty attaching to a statistical estimate, being the standard deviation of the sampling distribution of the statistic concerned, whether a mean or a regression coefficient, and it underwrites the construction of confidence intervals, which delimit the range of plausible values for the true population parameter given the observed sample; it is conventionally denoted σ and weighs heavily in the determination of statistical significance. Significance testing itself asks whether observed results are statistically significant or are instead the plausible offspring of chance: such tests assess the likelihood that an observed relationship between variables, or difference between groups, reflects random error rather than a true effect. The p-value, the probability of obtaining results at least as

extreme as those observed were the null hypothesis true, is the customary measure, and a p-value below 0.05 is conventionally taken to indicate that the results are unlikely to have arisen by chance.

Powerful though it is for the prediction of a continuous outcome, linear regression labours under limitations of which the analyst must remain aware. The assumption of linearity is the first and most confining: relationships in real-world data are frequently non-linear, and where they are, a non-linear specification, such as polynomial or logistic regression, may serve better. The method is, secondly, sensitive to outliers: observations lying far outside the range of the majority can, if undetected and unhandled, distort the fit materially, issuing in inaccurate predictions and misleading results. Thirdly, the model presumes the absence of multicollinearity among the predictors; where two or more independents are highly correlated with one another, the model's capacity for accurate prediction suffers. Finally, the residuals, the differences between predicted and observed values, are assumed to be normally distributed with constant variance, and violation of either assumption can compromise both predictions and the general performance of the model. Linear regression remains a useful and widely employed instrument, but the choice of model must answer to the character of the data and the research question in hand.

7.1.2 Support vector regression (SVR)

Support vector regression is a form of regression analysis that employs support vector machines (SVMs), supervised learning algorithms serviceable for classification and regression alike, to model and predict continuous variables. In SVR the object is the hyperplane in feature space that maximally separates the data points with the greatest attainable margin; the fitted hyperplane then furnishes predictions for new observations. The method commends itself for time-series analysis and prediction because it accommodates non-linear relationships in the data and performs creditably in high-dimensional settings; its application to a time series requires only that the model be trained upon the historical data before being turned to the prediction of future values.

Two differences from simple linear regression deserve notice. The first has already been intimated: SVR is able to model non-linear relations between dependent and independent variables, where linear regression is confined to linear ones, a versatility that tells particularly

when the data are ill-behaved or follow a non-linear pattern (T.S., 2020). The second concerns the cost function. Linear regression minimises the mean squared error between predicted and observed values; SVR employs instead a cost function based upon the distance between predicted and observed values, a construction that handles outliers and other extreme observations, which the MSE represents poorly, with greater composure. SVR is, in consequence, both more flexible and more robust to outliers than its linear counterpart.

7.1.3 Neural network

A neural network is a machine learning algorithm modelled upon the structure and function of the human brain: a large assembly of interconnected processing nodes, or neurons, organised into layers. Such networks are trained to their tasks by the adjustment of the weights on the connections between neurons, and once trained will generate predictions from new input data. They are well suited to time-series analysis and prediction, being capable of capturing complex non-linear relationships and of learning to recognise patterns in data as they unfold through time; their employment requires the design and training of a suitable network upon the historical series, after which the trained network may be set to predicting future values.

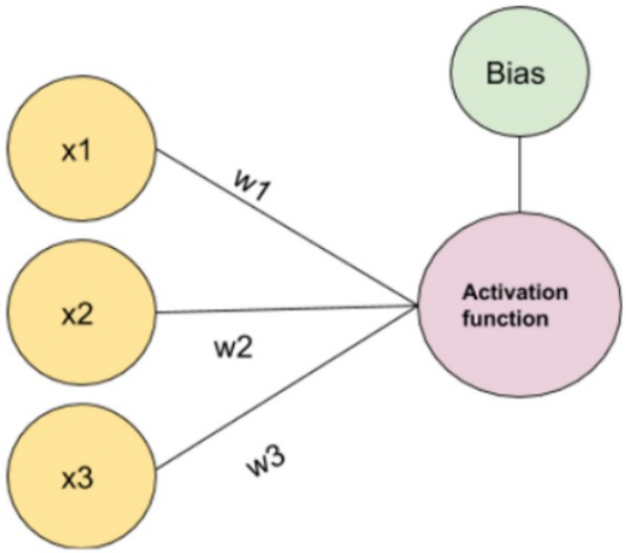


Diagram 1: The basic structure of a neuron

In the diagram the inputs to the neuron are denoted $x_1, x_2, \ldots, x_n$. Each is multiplied by its corresponding weight $w_1, w_2, \ldots, w_n$, the products are combined through a linear combination

function, and the result is passed through a transfer function, such as a sigmoid, to yield the neuron's final output. Both the weights and the transfer function are learned in training, so that the neuron comes to produce the desired output for a given input.

Within the neuron, inputs are combined and processed according to the weights and an aggregation function, a mathematical operation combining the several inputs into a single output value. The commonest aggregation function in neural networks is the linear combination, the sum of each input multiplied by its weight; for a neuron with three inputs x_1, x_2, x_3 and weights w_1, w_2, w_3:

$$\text{Output} = x_1 \times w_1 + x_2 \times w_2 + x_3 \times w_3$$

Alternative aggregation functions, among them sums of products, products of sums, maxima and minima, find employment in particular settings, and since the choice affects both the behaviour and the performance of the network, it belongs among the material decisions of network design. To the aggregated value an activation function is then applied: a mathematical function that fixes the range of possible outputs and, crucially, may introduce non-linearity, without which a network could not learn or model the complex, non-linear relationships that real data exhibit. Sigmoid, tanh and ReLU functions are among the common choices, and here too the selection bears significantly upon the network's behaviour and performance.

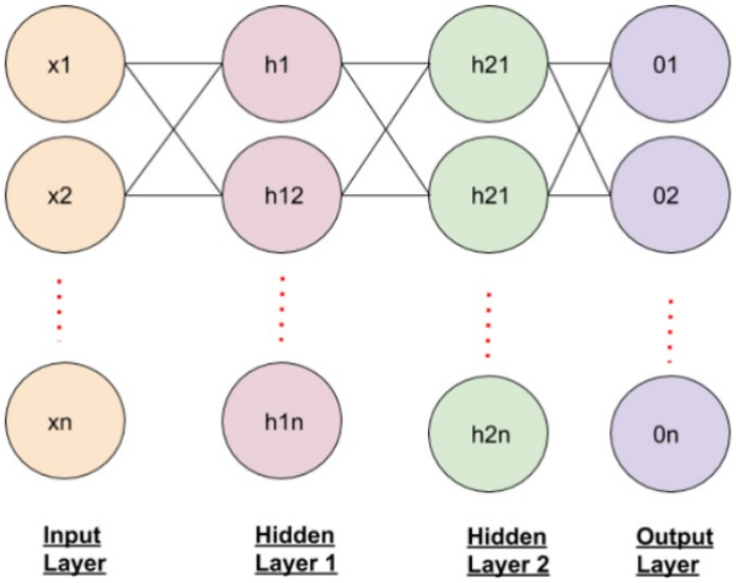


Diagram 2: Neural network model

The layers of a network are the successive sets of neurons stacked one upon another: an input layer, one or more hidden layers, and an output layer. The input layer receives the raw data and passes it to the first hidden layer; each hidden layer processes the output of its predecessor and passes the result onward. It is the hidden layers that lend the network its power, for without them a network could learn only linear relations between input and output, to the detriment of whole classes of problem; their addition permits the modelling of relationships of arbitrary complexity. The number of hidden layers, and of neurons within each, affects performance and behaviour alike, and these quantities are typically settled by experimentation and model tuning. The output layer, finally, produces the network's verdict: a prediction or classification for the input presented.

Training proceeds by the adjustment of the weights and biases of the network's neurons until the desired output is produced for a given input. The procedure is ordinarily one of supervised learning: the network is furnished with input-output pairs and must learn the function mapping the one to the other, the pairs being presented repeatedly while weights and biases are adjusted to minimise the error between predicted and true outputs. The adjustment process is known as backpropagation, the error being propagated backwards through the network to update the parameters. Training is computationally demanding and can be slow, but it is the indispensable condition of accurate prediction upon new and unseen data.

Applied to the present problem, a neural network may be employed to predict the future price of Bitcoin. Doing so requires first the collection of a dataset of historical Bitcoin prices, upon which a network is designed and trained; the network must learn the patterns and relationships within the series if its predictions are to carry conviction, and once trained it may be turned to forecasting. Its performance ought then to be evaluated upon a held-out test set, and its architecture and hyperparameters may require fine-tuning before that performance is satisfactory.

8.0 Data Analysis

8.1 Data Processing

Data preparation is an indispensable preliminary to regression analysis, bearing directly upon the accuracy and performance of any model built upon it. Proper preparation encompasses the gathering and combination of data from multiple sources; the cleaning and pre-processing of the data to remove errors, inconsistencies and missing values; and the organisation and restructuring of the result into a form suited to the intended use. To these may be added the selection and transformation of relevant features and variables, the normalisation and scaling of the data to improve model performance, and the division of the data into training, validation and test sets. Since it is preparation that assures the accuracy and suitability of the data, the care taken at this stage repays itself in the performance and reliability of the model, and ultimately in the usefulness of its predictions.

8.2 Data Gathering

The dataset employed in this research contains daily price information for Bitcoin over a fixed period, collected from the Yahoo Finance website (finance.yahoo.com/quote/BTC-USD/history). The data span very nearly eight years, from 1 January 2015 to 30 November 2022, and comprise seven attributes: the date together with the opening, high, low, closing and adjusted closing prices, and the traded volume. These attributes afford a detailed view of the asset’s performance across the period and lend themselves to a variety of uses, from the prediction of future prices to the analysis of market trends.

Date	Open	High	Low	Close	Adj Close	Volume
01/01/2015	320.434998	320.434998	314.002991	314.248993	314.248993	8036550
02/01/2015	314.07901	315.838989	313.565002	315.032013	315.032013	7860650
03/01/2015	314.846008	315.149994	281.082001	281.082001	281.082001	33054400
04/01/2015	281.145996	287.230011	257.612	264.195007	264.195007	55629100
05/01/2015	265.084015	278.341003	265.084015	274.473999	274.473999	43962800
06/01/2015	274.610992	287.553009	272.696014	286.188995	286.188995	23245700
07/01/2015	286.076996	298.753998	283.07901	294.337006	294.337006	24866800
08/01/2015	294.13501	294.13501	282.174988	283.348999	283.348999	19982500
09/01/2015	282.382996	291.114014	280.53299	290.40799	290.40799	18718600
10/01/2015	287.303009	288.127014	273.966003	274.79599	274.79599	15264300
11/01/2015	274.608002	279.638	265.039001	265.660004	265.660004	18200800
12/01/2015	266.145996	272.203003	265.200012	267.79599	267.79599	18880300
13/01/2015	267.394012	268.277008	219.906006	225.860992	225.860992	72843904
14/01/2015	223.893997	223.893997	171.509995	178.102997	178.102997	97638704
15/01/2015	176.897003	229.067001	176.897003	209.843994	209.843994	81773504
16/01/2015	209.070007	221.591003	199.770996	208.097	208.097	38421000
17/01/2015	207.834	211.731003	194.875	199.259995	199.259995	23469700
18/01/2015	200.050003	218.695007	194.505997	210.339005	210.339005	30085100
19/01/2015	211.470993	216.727997	207.317993	214.860992	214.860992	18658300
20/01/2015	212.906998	215.240997	205.153	211.315002	211.315002	24051100
21/01/2015	211.378006	227.787994	211.212006	226.897003	226.897003	29924600

Figure 2: Sample dataset

8.3 Regression Analysis

8.3.1 Linear Regression: Process and Results

The Python code employs a number of libraries, pandas and scikit-learn among them, to conduct a regression analysis upon the Bitcoin price dataset. The data are first loaded from the CSV file and sorted by date, and matplotlib is used to render a simple line graph of the price through time. The model is trained upon the x_Train and y_Train data, and its performance evaluated upon x_Test and y_Test through the score method, which returns a value between zero and one (higher values marking the better performance) that is printed to the console.

A new column, Prediction, is then created in the dataframe, containing the values of the Close column shifted by a specified number of rows; the displacement permits the data to be divided into training and test sets, the former serving to train the regression model and the latter to evaluate it. Four shift values of 30, 100, 200 and 365 rows are employed in this analysis to predict future prices at the corresponding horizons. The trained model is finally used to generate

predictions for the shifted values in the Prediction column, and these are plotted against the actual series so that the quality of the forecast may be inspected directly.

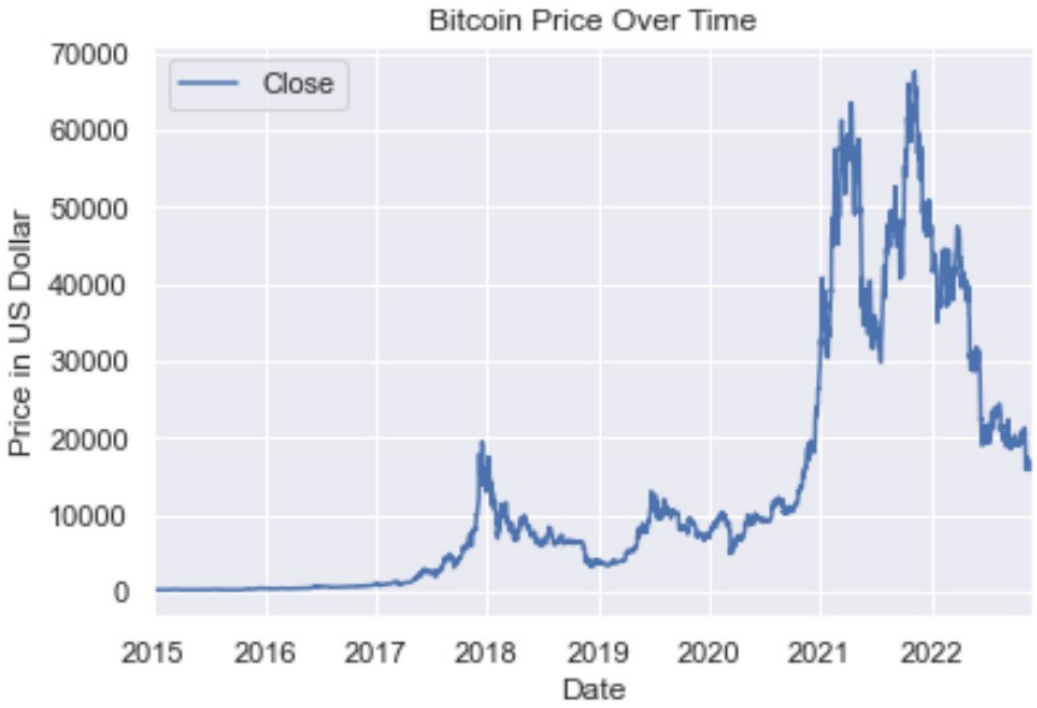


Figure 3: Bitcoin price on a time-series chart

The train_test_split function of scikit-learn divides the data into training and test sets; a linear regression model is constructed with the LinearRegression class and fitted to the training data; the fitted model is evaluated upon the test data and its performance printed; and predictions are generated and plotted as described.

The chart below sets Bitcoin's actual price against the price predicted by the regression model with a shift of 30 rows. At this horizon the linear regression model predicts the Bitcoin price with considerable accuracy.

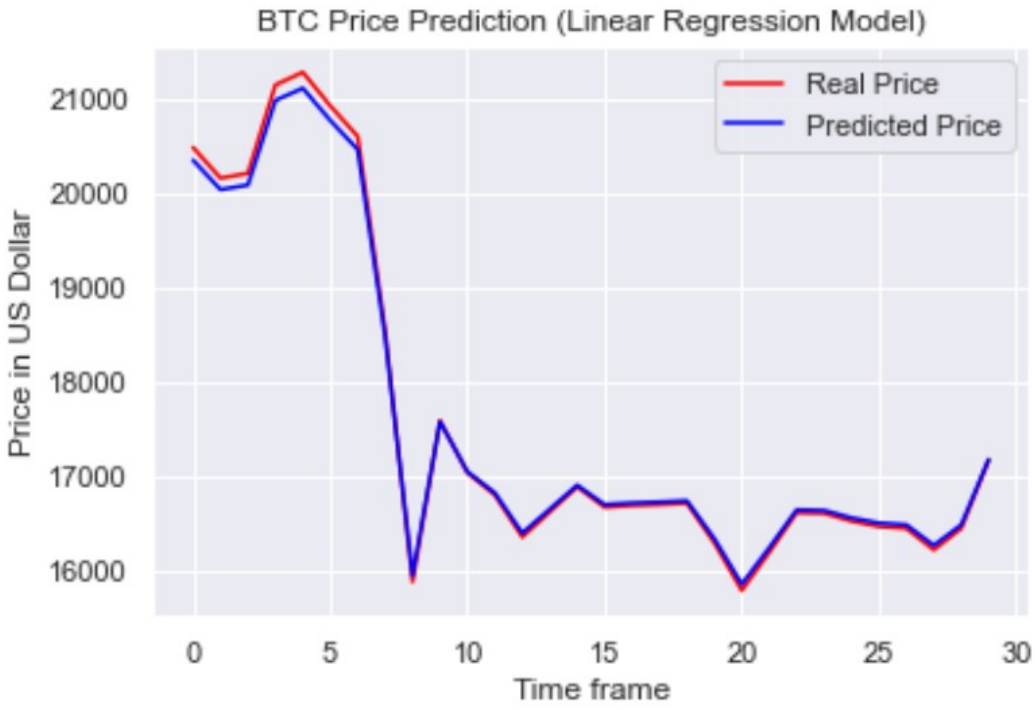


Figure 4: Bitcoin price prediction 1 (LRM)

With the shift extended to 100 rows, the prediction remains creditable, though its accuracy falls short of the preceding result.

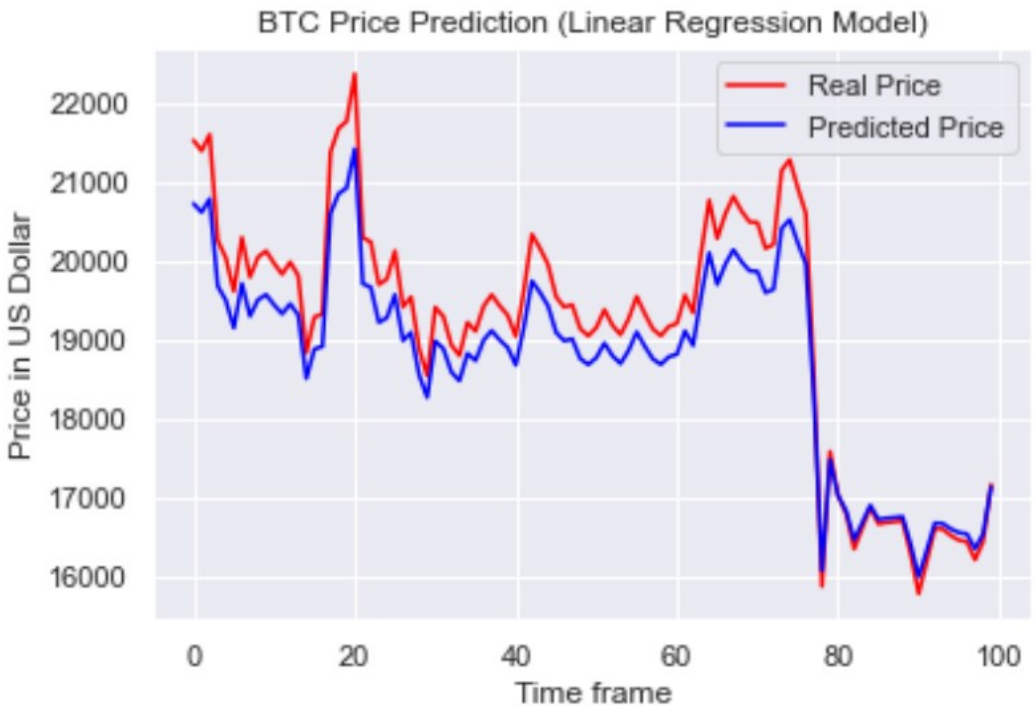


Figure 5: Bitcoin price prediction 2 (LRM)

At a shift of 200 rows the predicted price runs for the most part below the actual, though it exceeds it on occasion.

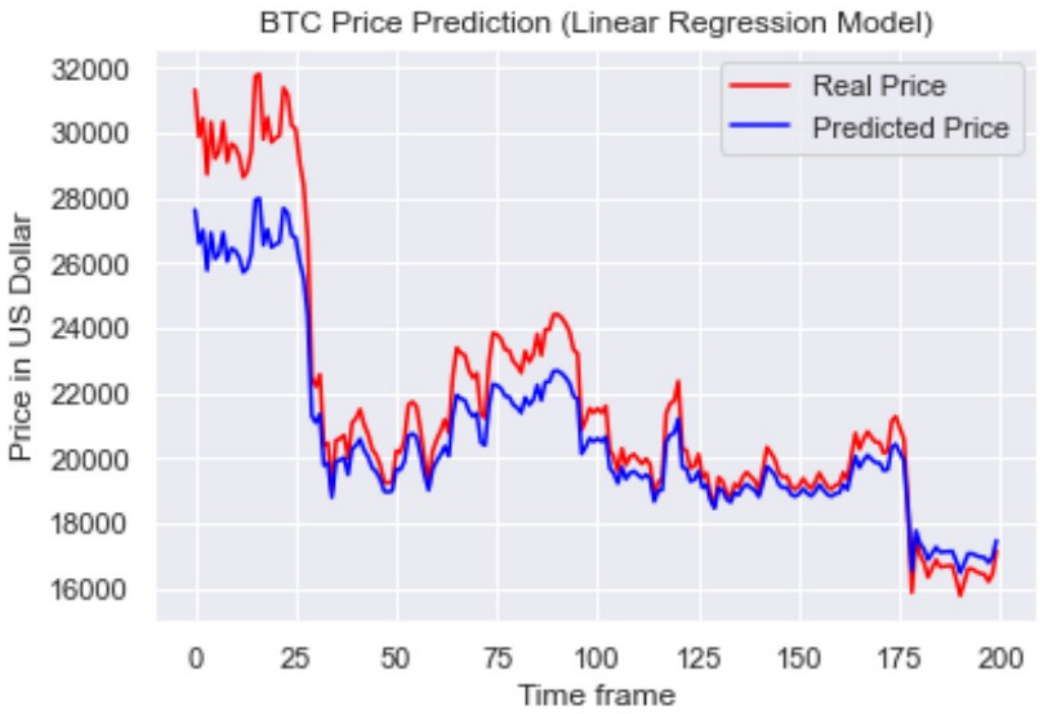


Figure 6: Bitcoin price prediction 3 (LRM)

At a shift of 365 rows the predicted price falls significantly below the actual, though the discrepancy narrows as the series progresses.

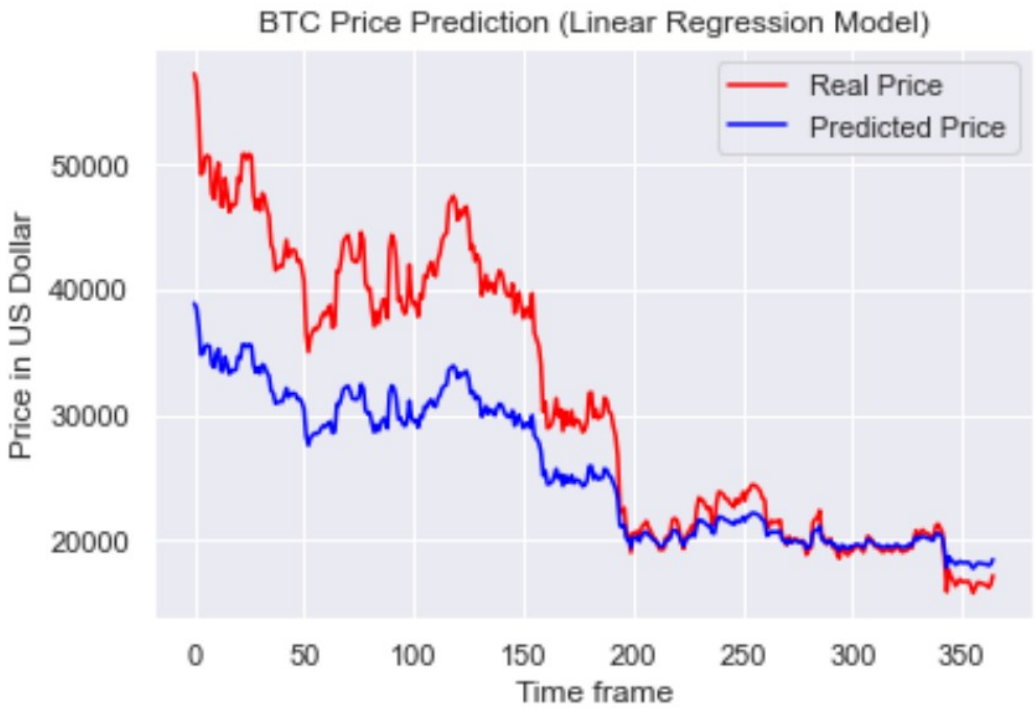


Figure 7: Bitcoin price prediction 4 (LRM)

8.3.2 Support Vector Regression (SVR): Process and Results

SVR, as a supervised learning algorithm applicable to regression, operates by locating the best possible hyperplane in feature space, that which separates the data most cleanly, and employing it to generate predictions upon new data. The code first loads the data and renders the familiar line graph; a Prediction column of shifted Close values is created; the data are divided into training and test sets; and two SVR models are constructed, the one with a radial basis function (RBF) kernel and the other with a polynomial kernel, each trained upon the training data and evaluated upon the test data. The trained models are then used to predict the shifted values, and the predictions plotted against the actual series.

The two kernels merit brief explanation. A radial basis function is a mathematical function, employed across several fields including machine learning, whose output depends upon the distance between the input and a fixed point; for a multidimensional input vector, the output depends upon the distance between that vector and the fixed point. RBFs commonly serve as bases for the construction of other functions, and appear in clustering algorithms and other techniques wherever the similarity of points must be measured. The polynomial kernel, for its

part, is a kernel function computing the dot product of two vectors raised to a given power; it finds frequent use in SVMs where the data are not linearly separable, that is, incapable of division into classes by a single line; in such circumstances it permits the model to learn the more complex decision boundary the data demand.

The SVM was trained upon historical Bitcoin prices and set to predicting future ones under both kernel functions, the predictions being plotted against actual prices for the evaluation of performance.

The chart below shows actual and predicted prices under the SVM with a shift of 30 rows. The model employing the RBF kernel outperforms its polynomial counterpart; yet both under-predict the actual price, suggesting that the SVM may be ill-suited to the prediction of Bitcoin prices, a conclusion that must remain provisional absent fuller information as to the data and model parameters employed.

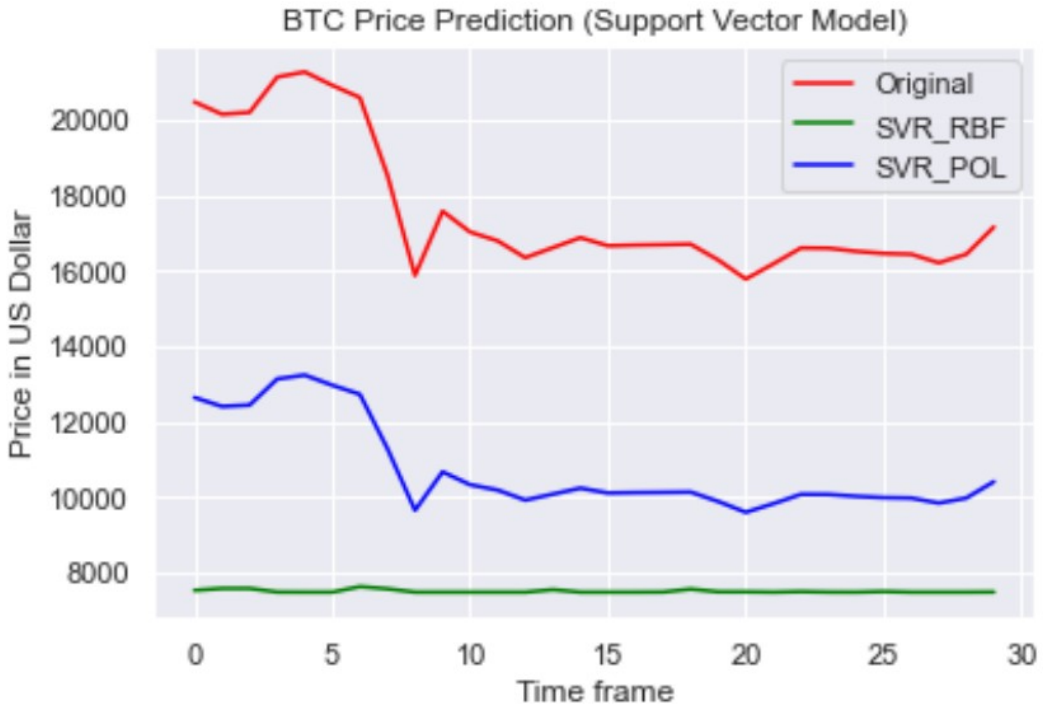


Figure 8: Bitcoin price prediction 1 (SVM)

At a shift of 100 rows the pattern repeats: the RBF kernel again proves the stronger, and both models again under-predict, with the same caveat as to the provisional character of the conclusion.

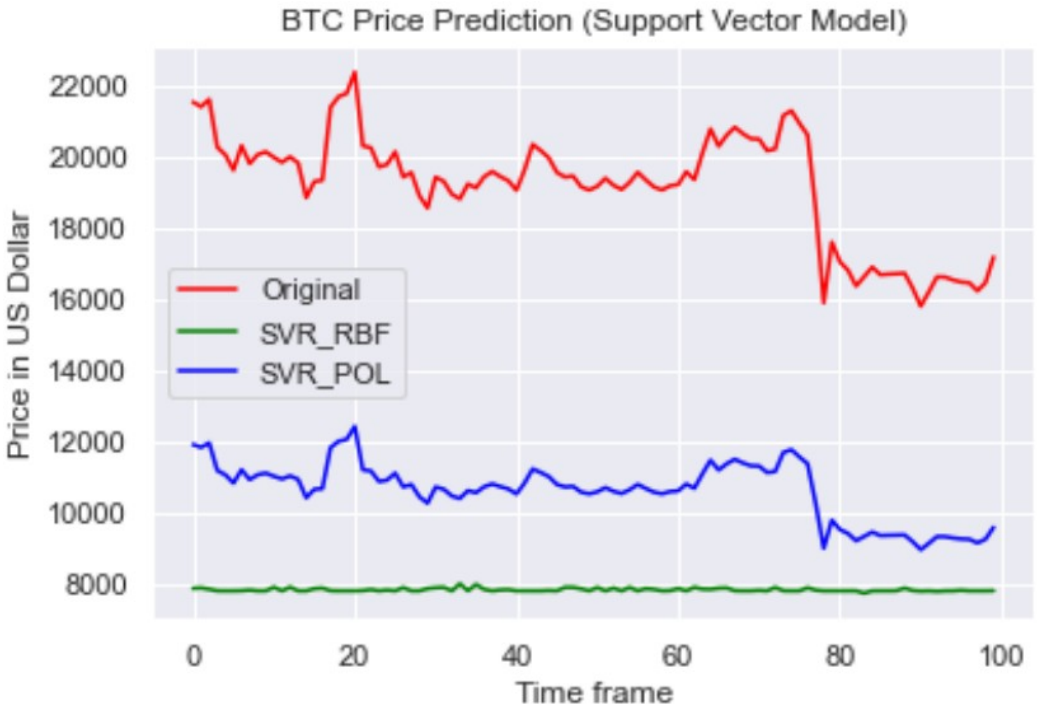


Figure 9: Bitcoin price prediction 2 (SVM)

The result at 200 rows is materially unchanged: the RBF kernel retains its advantage, and the under-prediction persists.

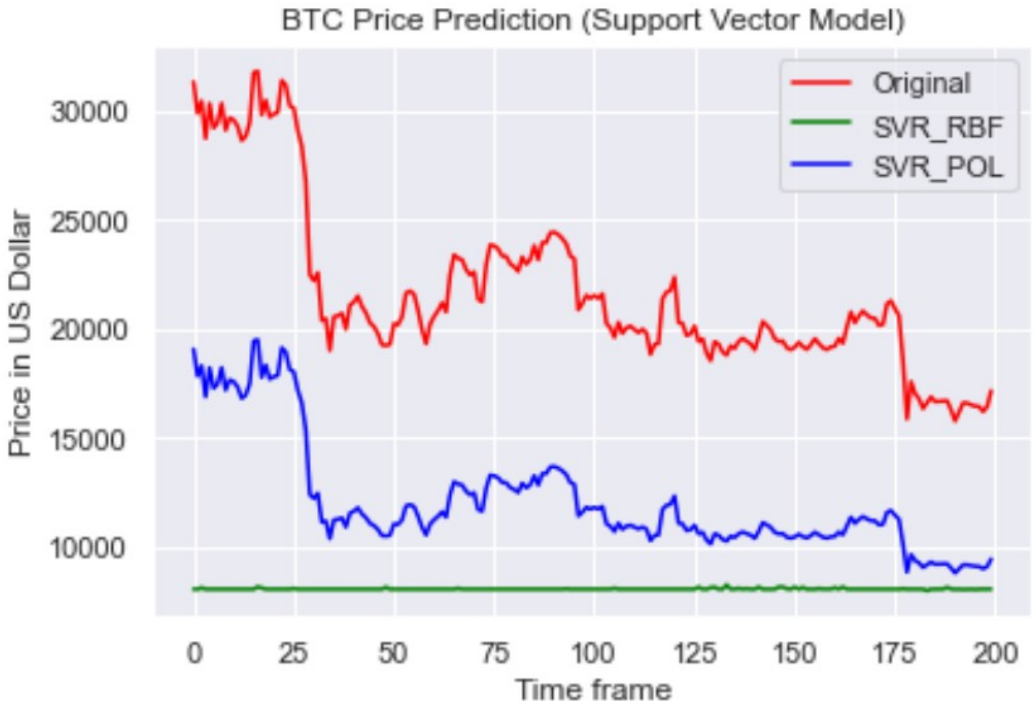


Figure 10: Bitcoin price prediction 3 (SVM)

At 365 rows the RBF kernel once more outperforms the polynomial, and the prediction improves upon those obtained at shorter shifts; yet both models continue to under-predict the actual price, and the reservation entered above, that certainty would require fuller knowledge of the data and parameters, applies with undiminished force.

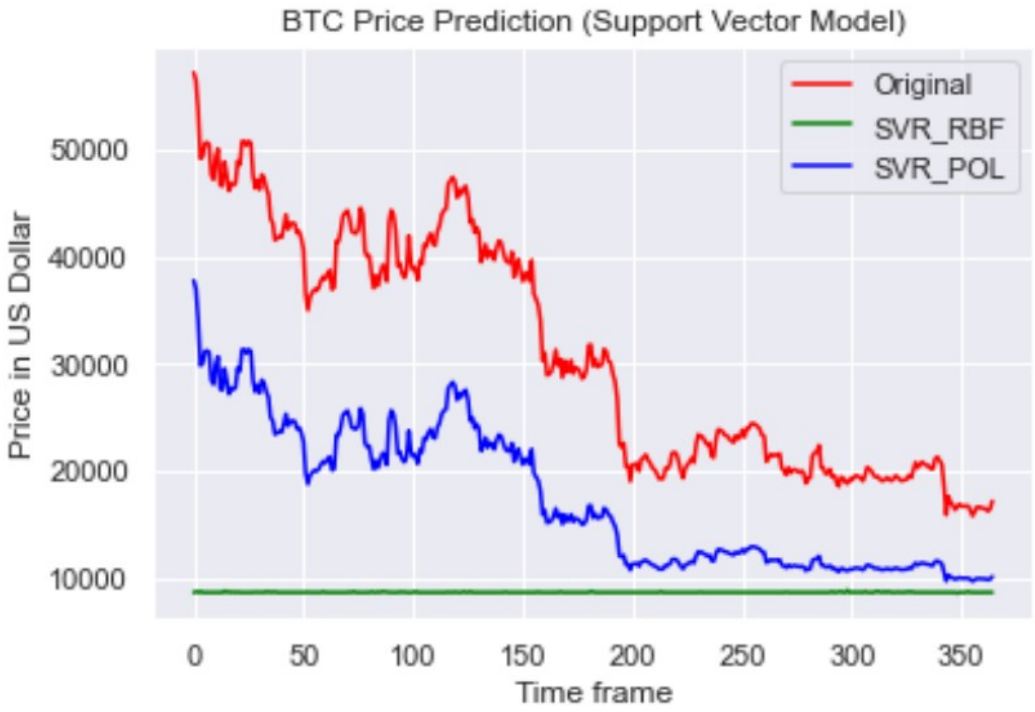


Figure 11: Bitcoin price prediction 4 (SVM)

8.3.3 Neural Network Regression: Process and Results

The Python code employs a neural network, a machine learning model inspired by the structure and function of the human brain and consisting of successive layers of interconnected neurons capable of processing complex data, to generate predictions upon the Bitcoin price series. The data are loaded and graphed as before; the Prediction column of shifted Close values is created; and the data are divided into training and test sets.

The network itself is defined with the Sequential class of tensorflow.keras, and comprises an input layer, several hidden layers and an output layer. The hidden layers employ the ReLU activation function, equipping the network to learn non-linear relationships within the data, while the output layer employs the linear activation function, the network's output being a continuous value. The model is compiled and trained upon the training data, whereupon it is used to predict the shifted values of the Prediction column and the predictions are plotted against the actual series.

The chart below shows actual and predicted prices under the neural network model with a shift of 30 rows. The model outperforms the SVR in the prediction of the Bitcoin price, yet it too

under-predicts, an indication that further adjustment might be required before its forecasts could be preferred.

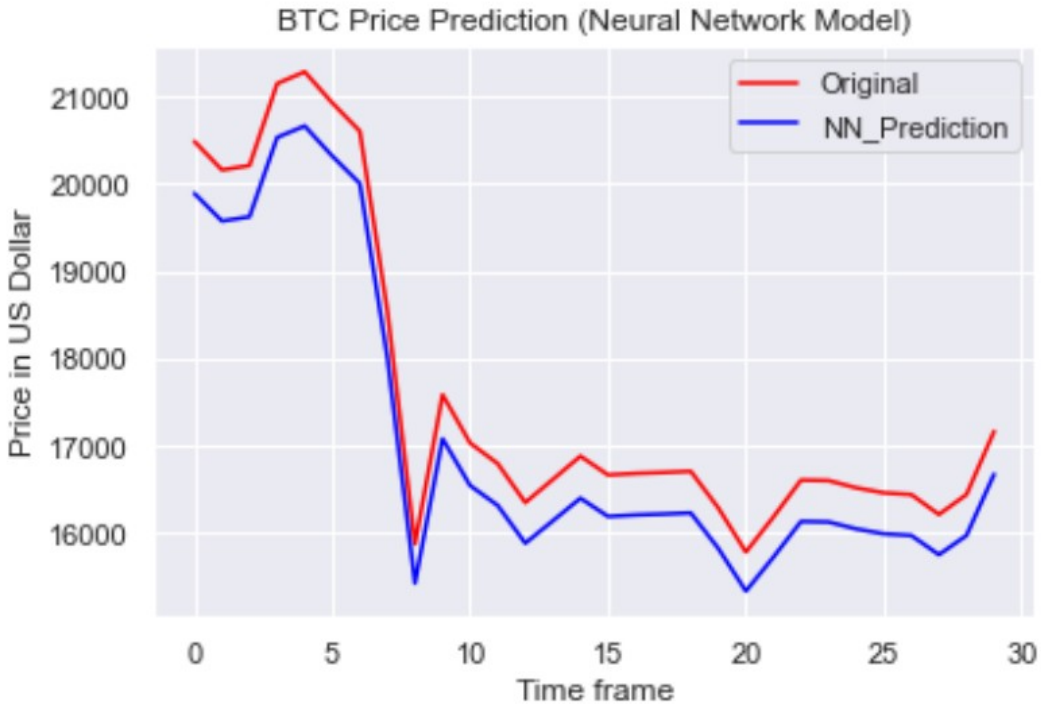


Figure 12: Bitcoin price prediction 1 (NN)

At a shift of 100 rows the under-prediction persists and widens: the gap between actual and predicted price is more pronounced than at the 30-row horizon.

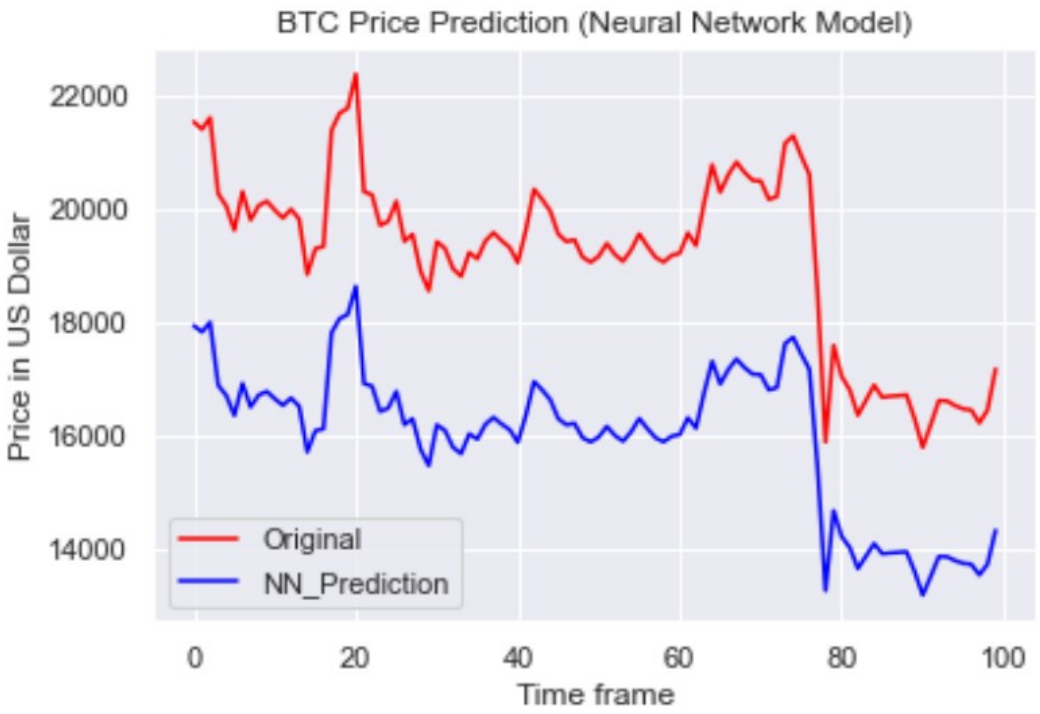


Figure 13: Bitcoin price prediction 2 (NN)

The same pattern obtains at 200 rows, the discrepancy between actual and predicted price remaining substantial relative to the shortest horizon.

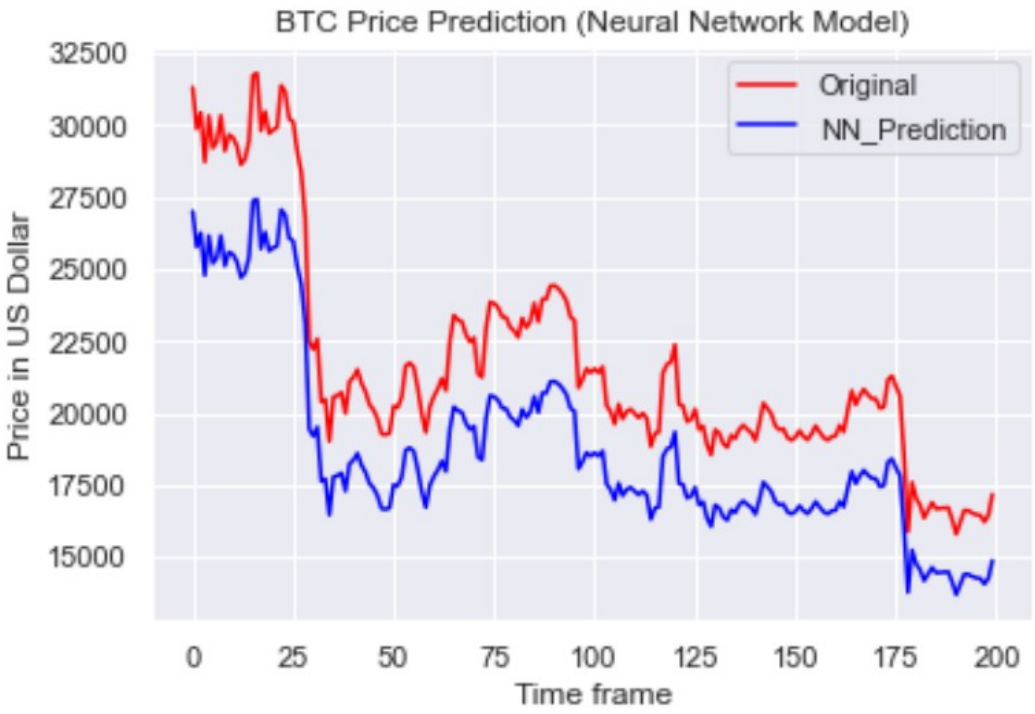


Figure 14: Bitcoin price prediction 3 (NN)

At 365 rows the model again under-predicts, and the divergence from the actual series remains significant when set against the results at 30 rows.

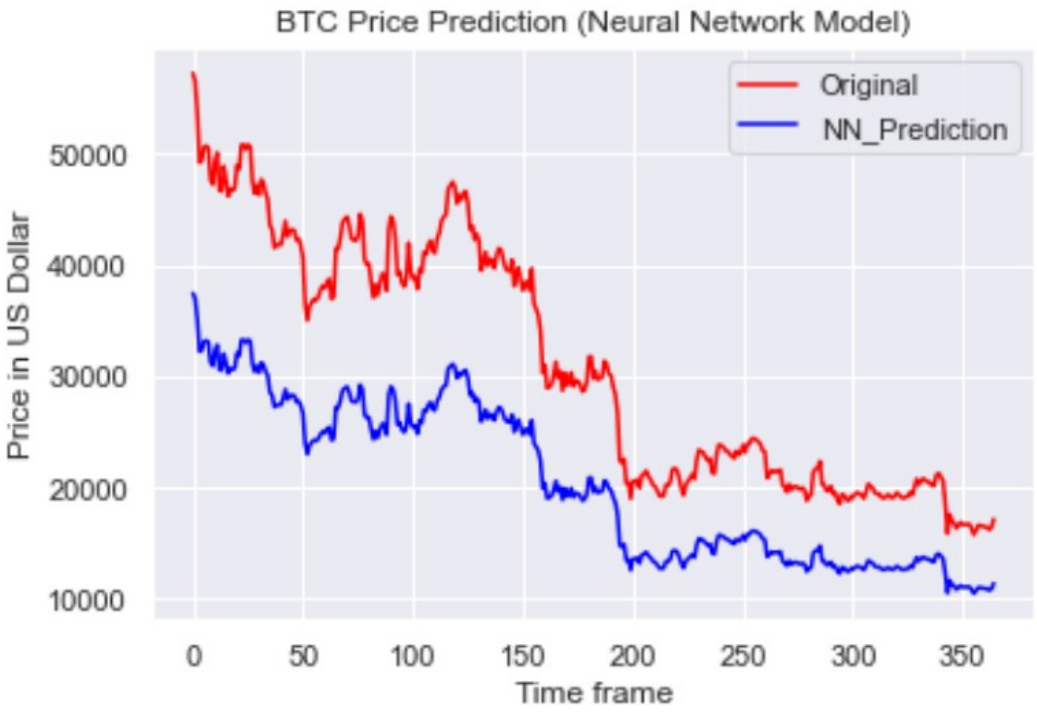


Figure 15: Bitcoin price prediction 4 (NN)

Upon the evidence of the regression analysis taken as a whole, the linear regression model proves the more accurate of the three, outperforming both the support vector regression and the neural network. The neural network nonetheless performed creditably, and its accuracy might well be improved by the adjustment of certain parameters within the model.

8.4 Correlation

8.4.1 BTC Correlation with Other Cryptoassets and Stocks

The Python code imports several libraries for the retrieval and plotting of financial data: pandas_datareader for the collection of historical data on a list of cryptocurrencies, matplotlib for plotting, datetime for the specification of the date range, and mplfinance and seaborn for the further customisation of the charts. The currency and metric are defined at the outset, the currency being set to USD and the metric to Close so that closing prices are retrieved in US dollars; the start and end dates are then specified, the list of cryptocurrencies drawn up, data fetched for each and the several series combined into a single DataFrame. The price plot is

customised with a logarithmically scaled y-axis and a legend identifying each asset, and displayed.

For the correlation analysis proper, the procedure differs in one material respect: daily price changes are employed rather than closing price levels. The code computes the percentage change for each cryptocurrency, derives the correlation matrix by means of the pandas corr method, and renders the result as a heatmap through the seaborn library. The identical method is applied in the later sections to establish the correlation of BTC and ETH with metaverse and NFT cryptoassets and with traditional assets.

8.4.2 Reading the Correlation Matrices

The resulting heatmaps display the correlation between the daily price changes of the specified assets, darker cells denoting the stronger correlations and lighter cells the weaker, with the figure within each cell recording the Pearson correlation coefficient for the pair concerned. The Pearson coefficient measures the linear correlation between two variables and ranges from −1 to 1: a value of −1 marks a strong negative linear relationship, zero the absence of any linear relationship, and 1 a strong positive one. It is the standard instrument for establishing whether, and to what degree, two variables move together.

8.4.3 Correlation between Bitcoin and Other Cryptoassets

For this analysis six further large-capitalisation cryptocurrencies are set alongside Bitcoin: Ethereum (ETH), Ripple (XRP), Polkadot (DOT), Solana (SOL), Chainlink (LINK) and Cardano (ADA). Five datasets of daily prices are examined, covering respectively the periods 1 January 2018 to 30 November 2022; 1 January 2021 to 31 January 2022; 1 February 2022 to 31 May 2022; 1 June 2022 to 30 November 2022; and 1 October 2021 to 30 November 2021.

1 January 2018 to 30 November 2022. The chart below presents the prices of the selected cryptocurrencies, Bitcoin included, on a logarithmic scale, using daily data for the full period.

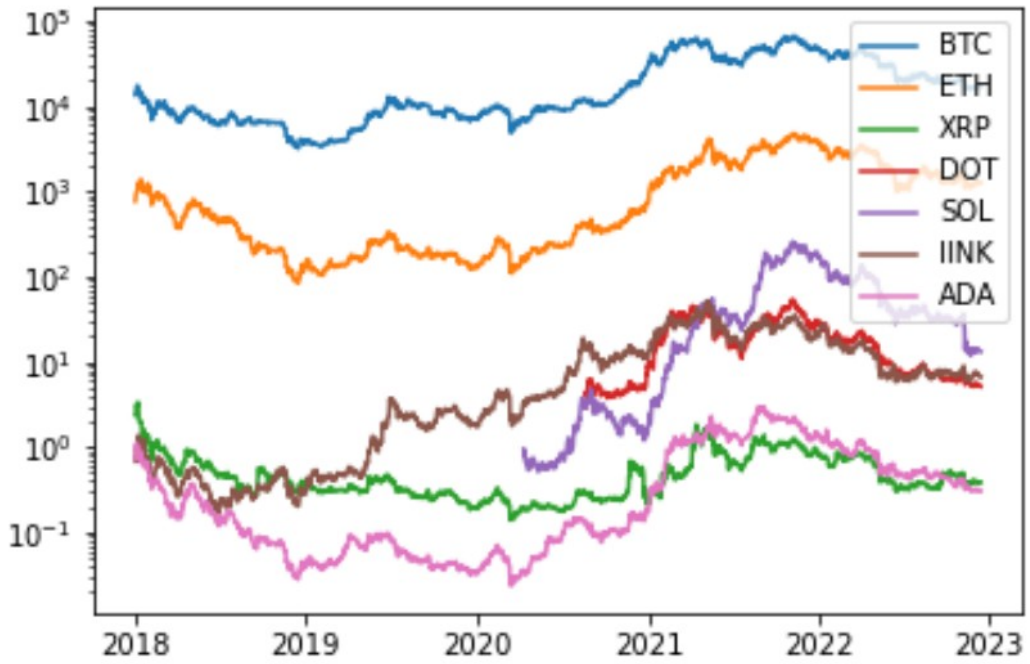


Figure 16: Cryptoasset prices on a logarithmic scale

The correlations among the seven assets appear in the table below.

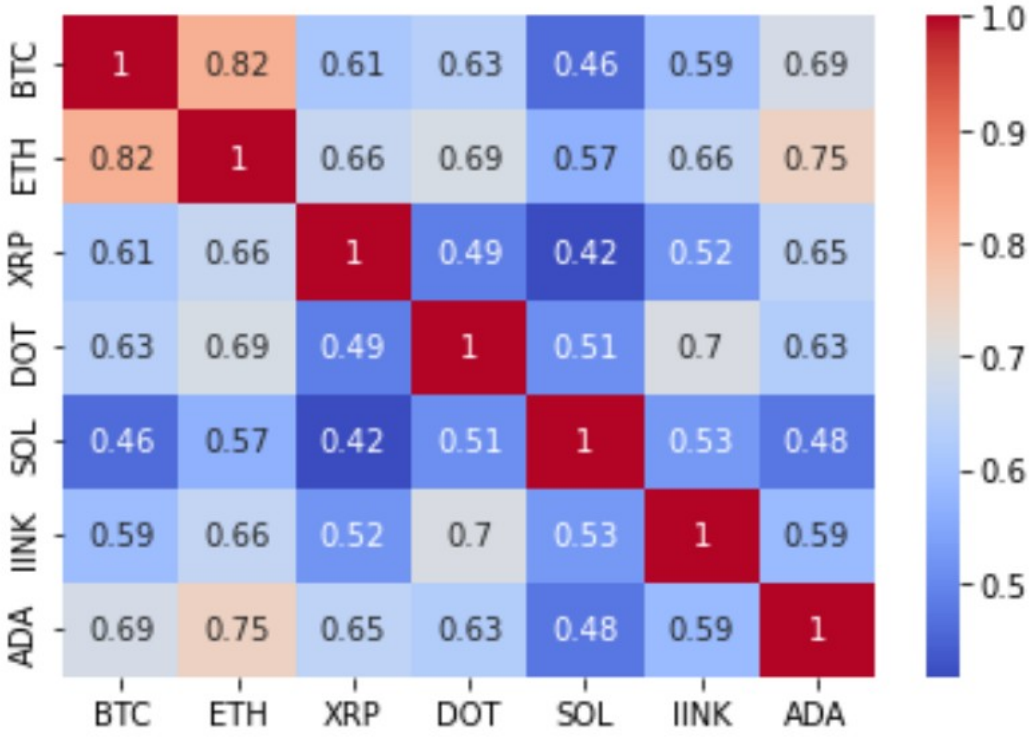


Table 3: Pearson correlation coefficients between cryptoassets

Across the full period the Pearson coefficient between BTC and ETH is the highest observed, at 0.82. That between ETH and ADA is likewise elevated, at 0.75; this is unsurprising, perhaps, since both platforms offer the creation of decentralised applications upon their blockchains. The coefficients across all pairs range from 0.42 to 0.82.

1 January 2021 to 31 January 2022. The chart below presents the same assets on a logarithmic scale for the thirteen months to January 2022.

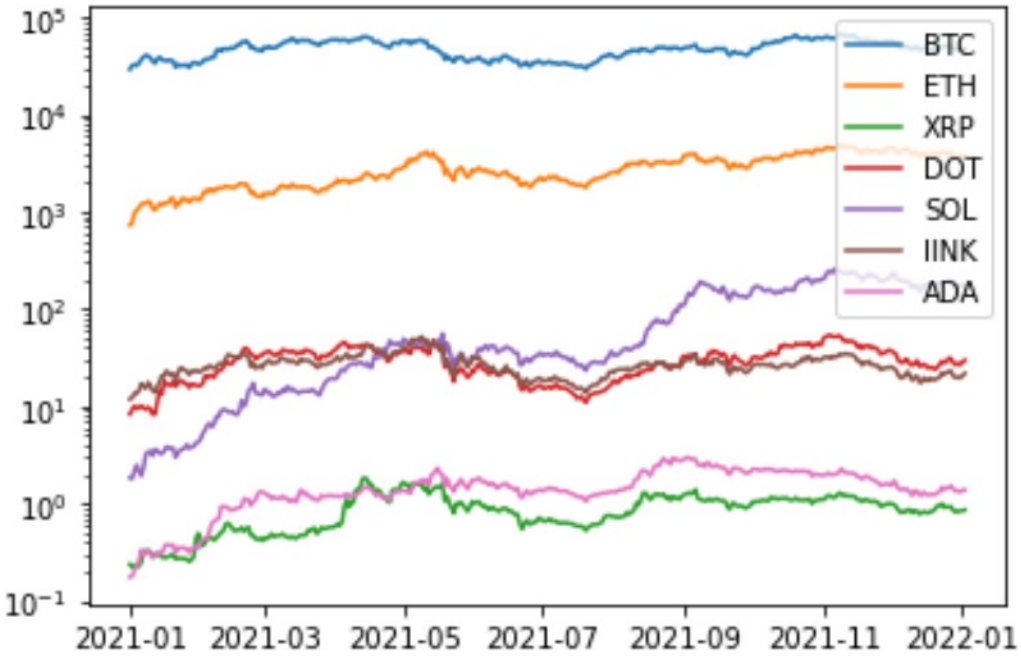


Figure 17: Cryptoasset prices on a logarithmic scale

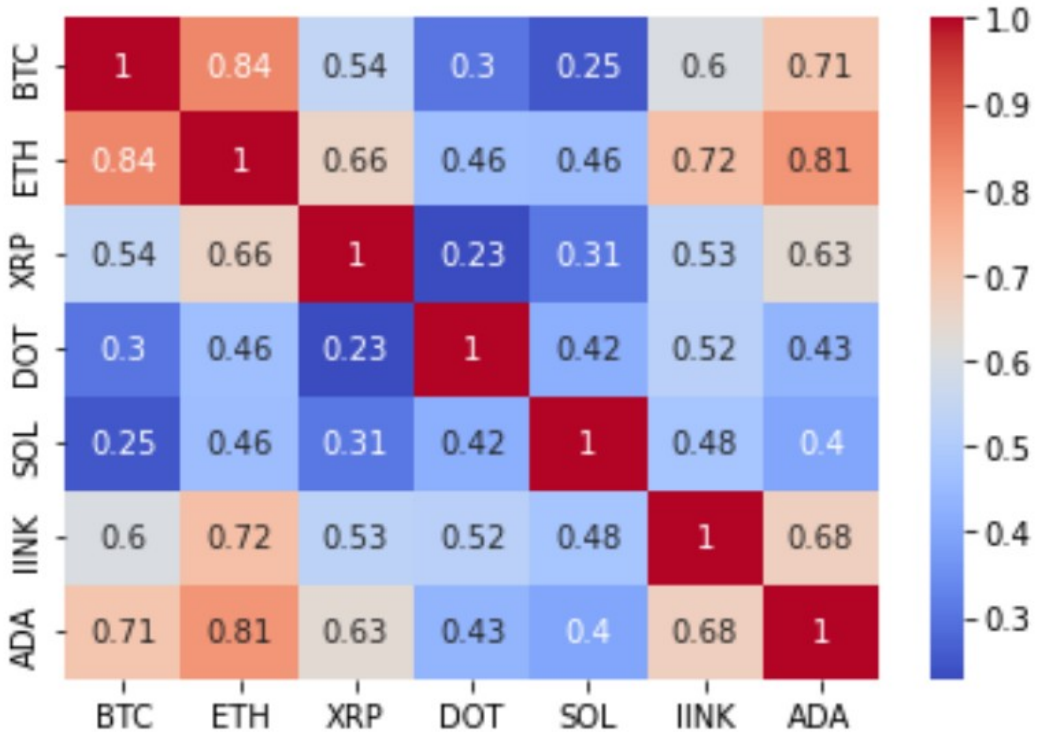


Table 4: Pearson correlation coefficients between cryptoassets

Here again the pairing of BTC with ETH heads the table, at 0.84, with ETH and ADA close behind at 0.81, the affinity between the two smart-contract platforms persisting. The full range of coefficients runs from 0.23 to 0.84.

1 February 2022 to 31 May 2022. The chart below covers the four months from February to May 2022.

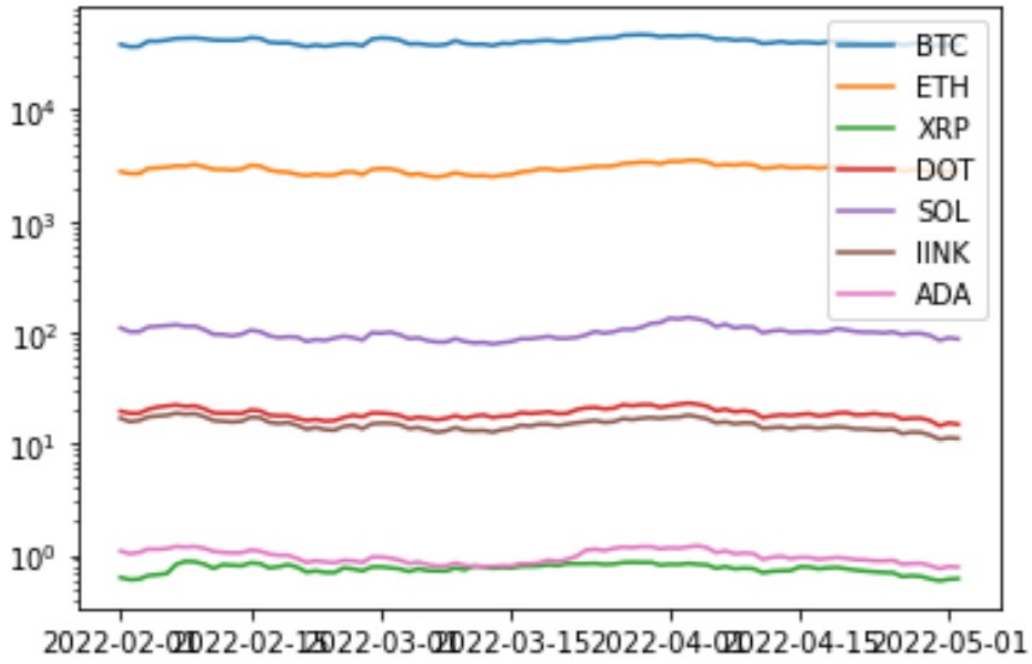


Figure 18: Cryptoasset prices on a logarithmic scale

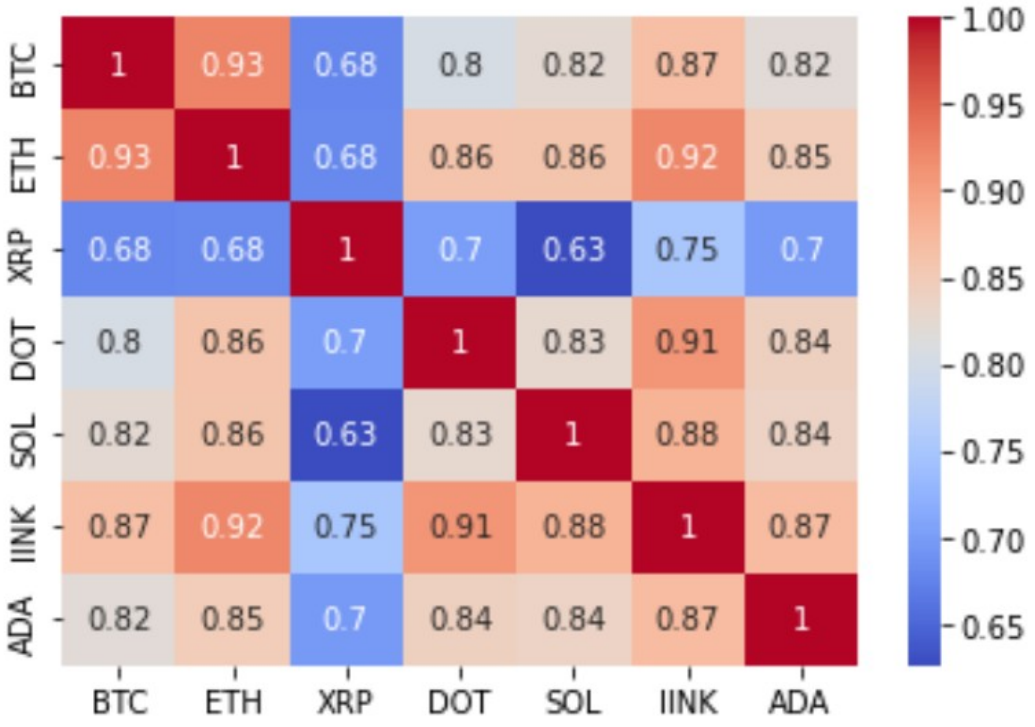


Table 5: Pearson correlation coefficients between cryptoassets

In this window the coefficient between BTC and ETH reaches 0.93, its highest reading across the periods examined, with ETH paired with LINK second at 0.92, and ETH with ADA also elevated

at 0.85. The coefficients range from 0.63 to 0.93, a general tightening of co-movement across the market.

1 June 2022 to 30 November 2022. The chart below covers the final six months of the sample.

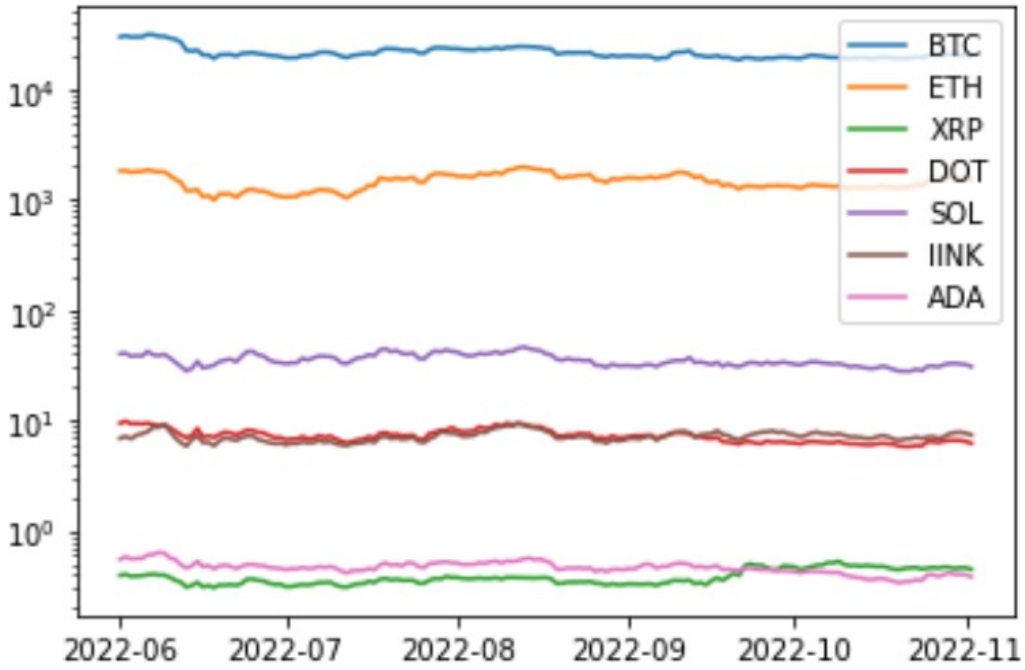


Figure 19: Cryptoasset prices on a logarithmic scale

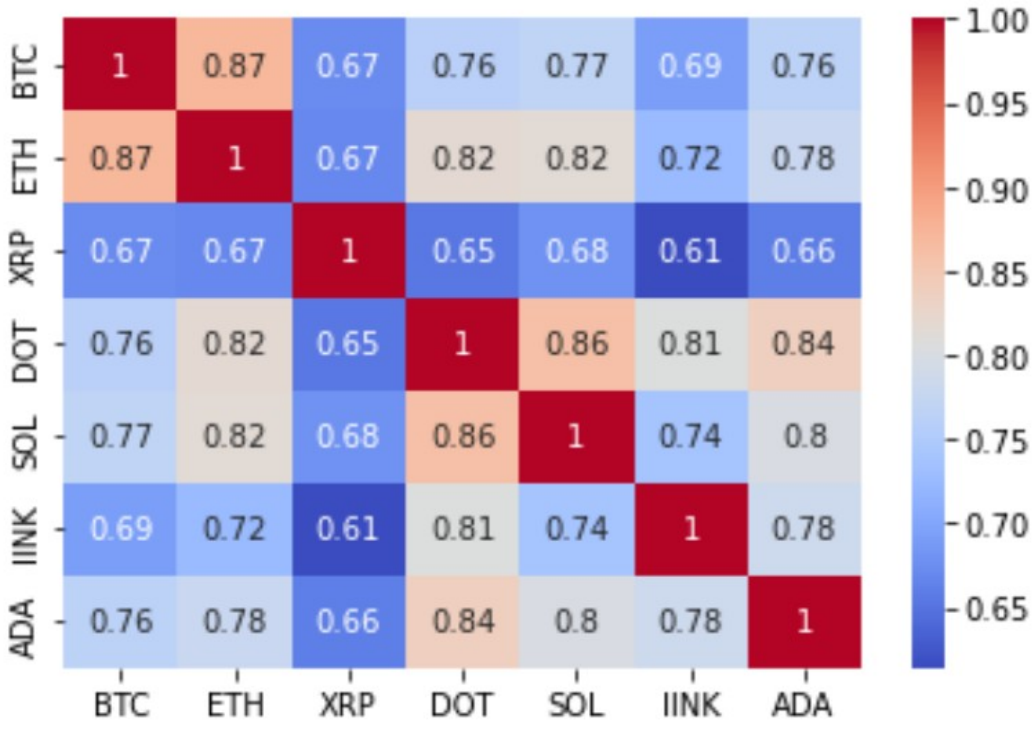


Table 6: Pearson correlation coefficients between cryptoassets

The BTC and ETH pair again stands highest, at 0.87, followed by DOT with SOL at 0.86, and ETH with ADA at 0.78. The coefficients range from 0.61 to 0.87.

1 October 2021 to 30 November 2021. Bitcoin attained its all-time high in the middle of November 2021, and this dataset was selected precisely to examine correlation during the peak of the bull market (time.com/nextadvisor/investing/cryptocurrency/price/bitcoin-btc/). The chart below presents the daily prices for those two months on a logarithmic scale.

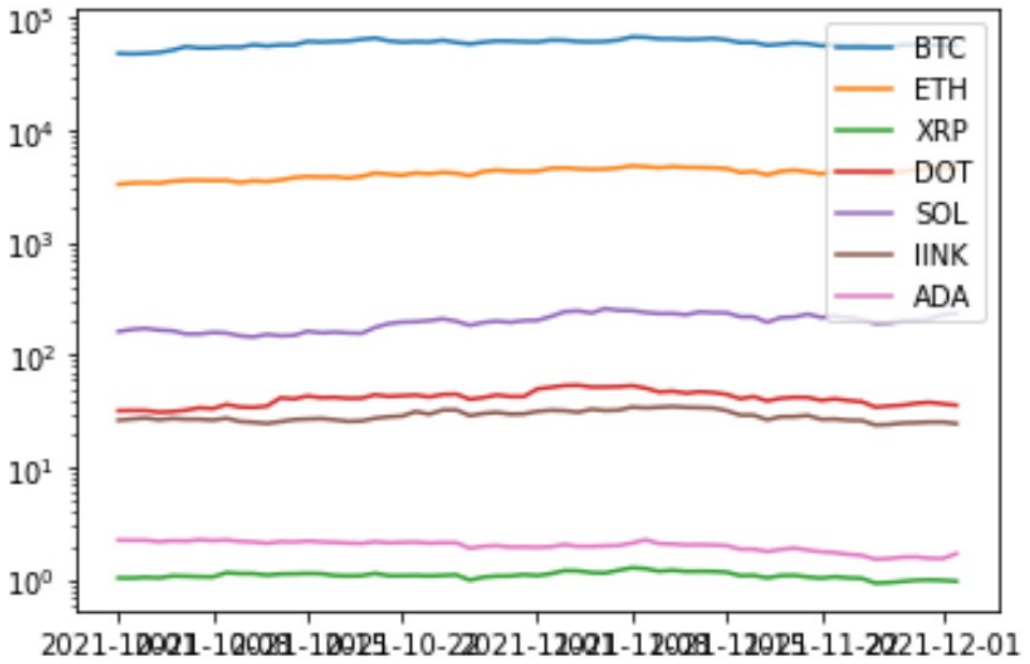


Figure 20: Cryptoasset prices on a logarithmic scale

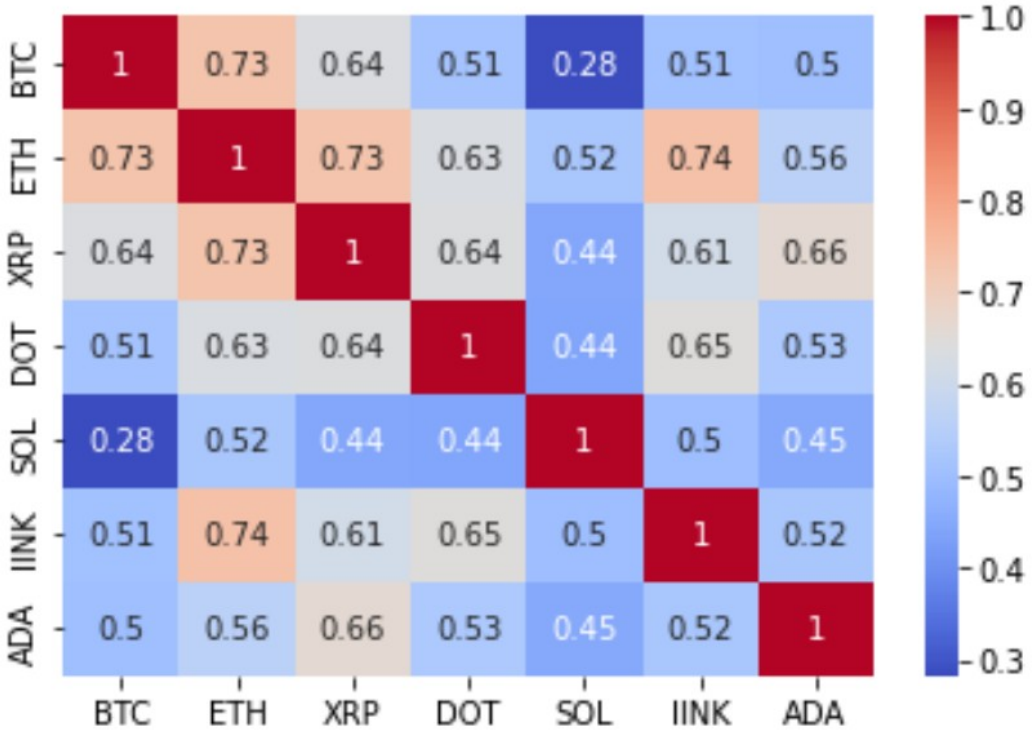


Table 7: Pearson correlation coefficients between cryptoassets

The picture at the market's peak differs markedly from the other windows. Co-movement weakens across the board, the coefficients ranging from 0.28 to 0.74: the ETH and LINK pair records the highest reading, at 0.74, marginally exceeding BTC and ETH at 0.73, while the pairing of BTC with SOL falls as low as 0.28. The evidence suggests that in the euphoria of a bull-market peak the constituent assets of the market part company to a degree not observed in calmer or declining conditions.

8.4.4 BTC and ETH Correlation with Traditional Assets

For this analysis five large-capitalisation traditional equities are set alongside Bitcoin and Ethereum, namely Tesla (TSLA), Apple (AAPL), Amazon (AMZN), Alphabet (GOOGL) and Meta Platforms (META), using daily closing prices between 1 January 2020 and 30 November 2022. The chart below presents the assembled series on a logarithmic scale.

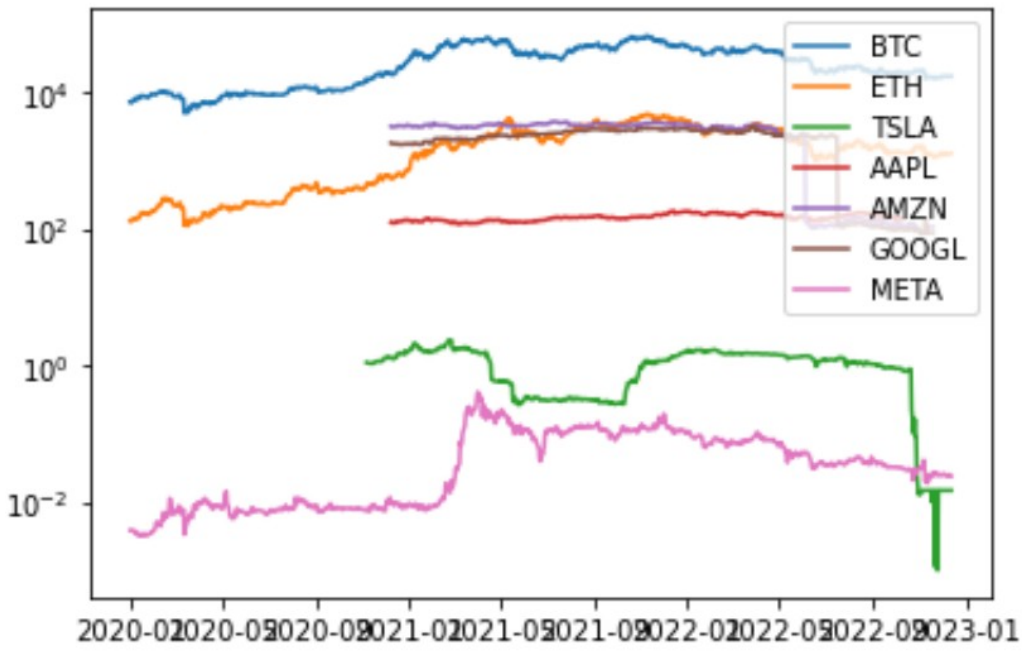


Figure 21: Cryptoasset and equity prices on a logarithmic scale

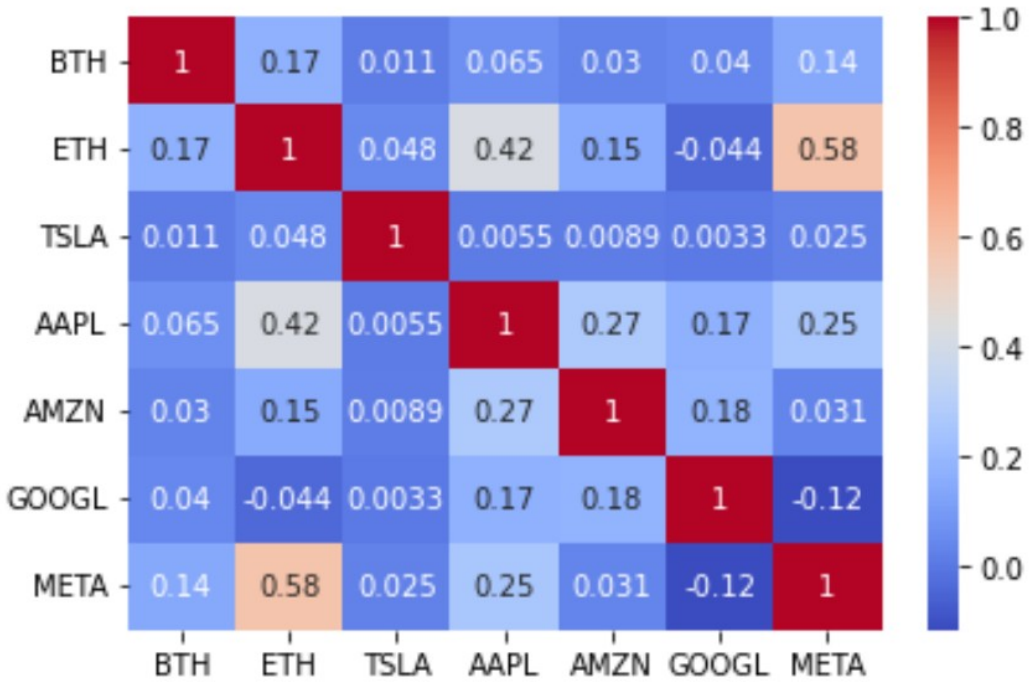


Table 8: Pearson correlation coefficients between cryptoassets and equities

The strongest association between the two asset classes is that of ETH with META, at 0.58, followed by ETH with AAPL at 0.42. Bitcoin's correlations with the selected equities are uniformly weak, none exceeding 0.14, so that on this evidence it is Ethereum, rather than Bitcoin, that exhibits the closer connection with large-capitalisation technology stocks.

8.4.5 BTC and ETH Correlation with NFT/Metaverse Cryptocurrencies

The final analysis sets Bitcoin and Ethereum alongside five prominent NFT and metaverse cryptocurrencies: ApeCoin (APE), Flow (FLOW), The Sandbox (SAND), Axie Infinity (AXS) and Decentraland (MANA). Two datasets of daily prices are examined, covering 1 January 2020 to 30 June 2021 and 1 July 2021 to 30 November 2022 respectively.

1 January 2020 to 30 June 2021. The chart below presents the NFT and metaverse assets together with Bitcoin and Ethereum on a logarithmic scale.

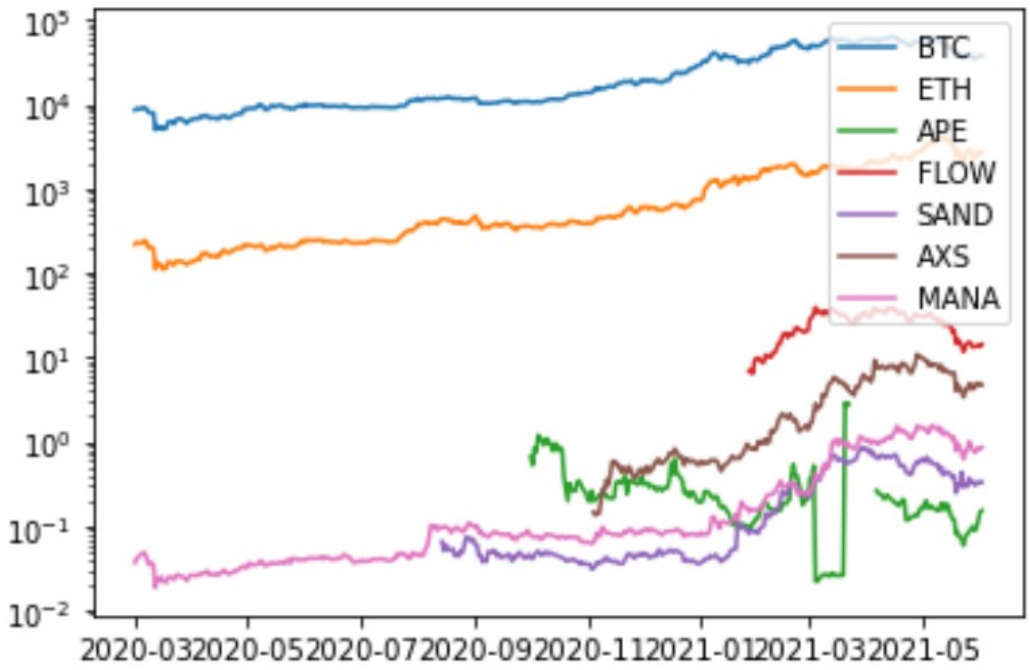


Figure 22: Cryptoasset prices on a logarithmic scale

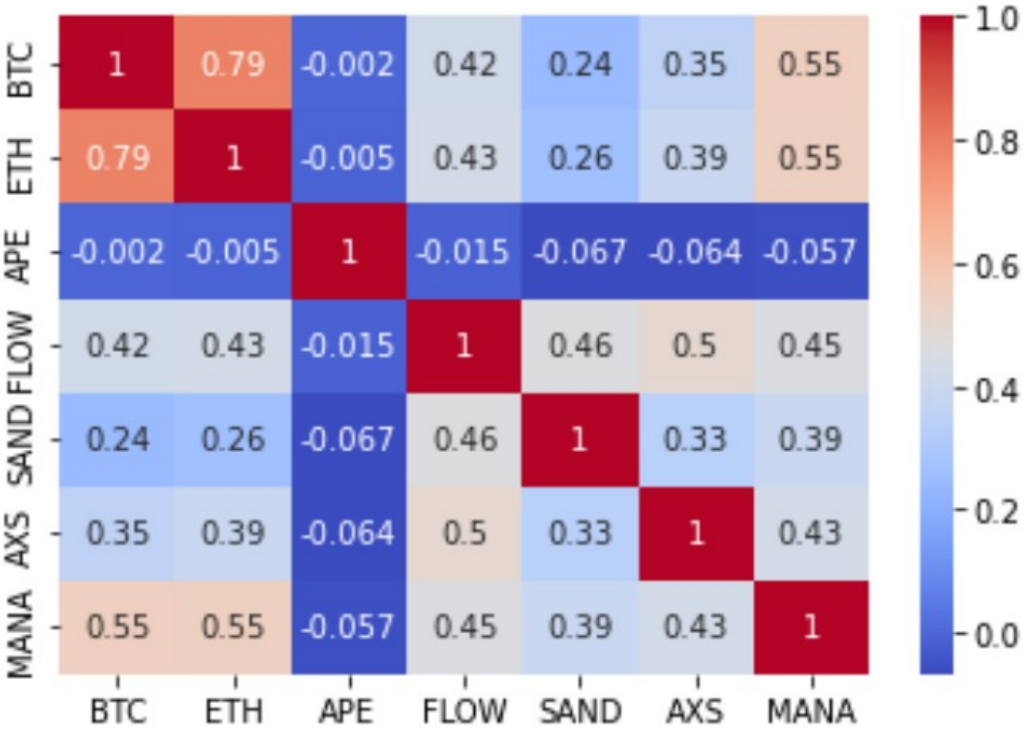


Table 9: Pearson correlation coefficients between cryptoassets

The highest coefficient in this window is that between ETH and MANA, at 0.55, a result consonant with MANA's character as a metaverse project built upon the Ethereum blockchain network. APE, by contrast, exhibits a negative linear relationship with both BTC and ETH.

1 July 2021 to 30 November 2022. The chart below presents the same assets over the later period.

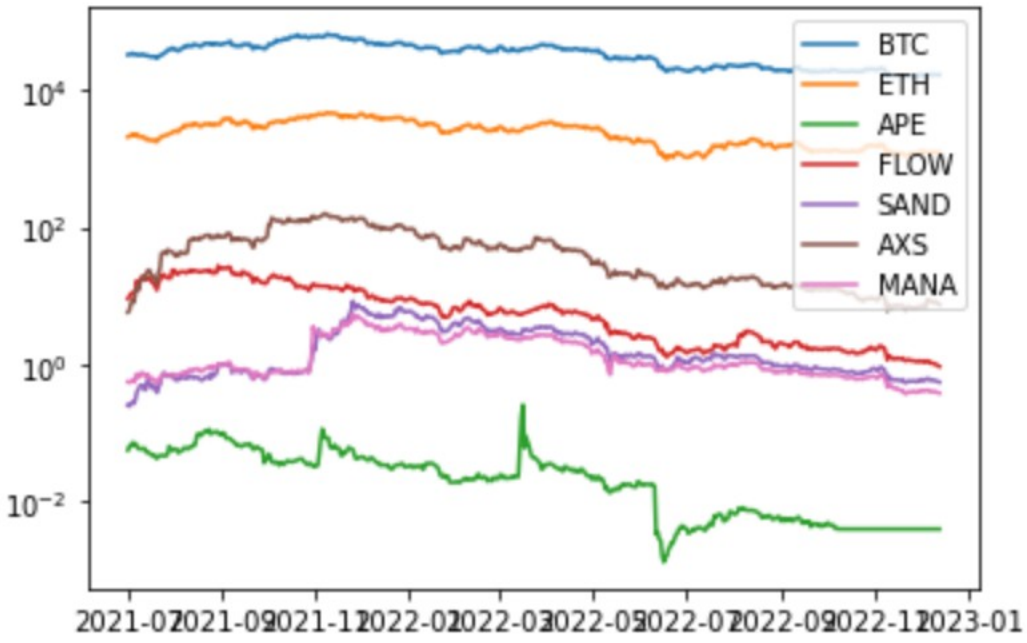


Figure 23: Cryptoasset prices on a logarithmic scale

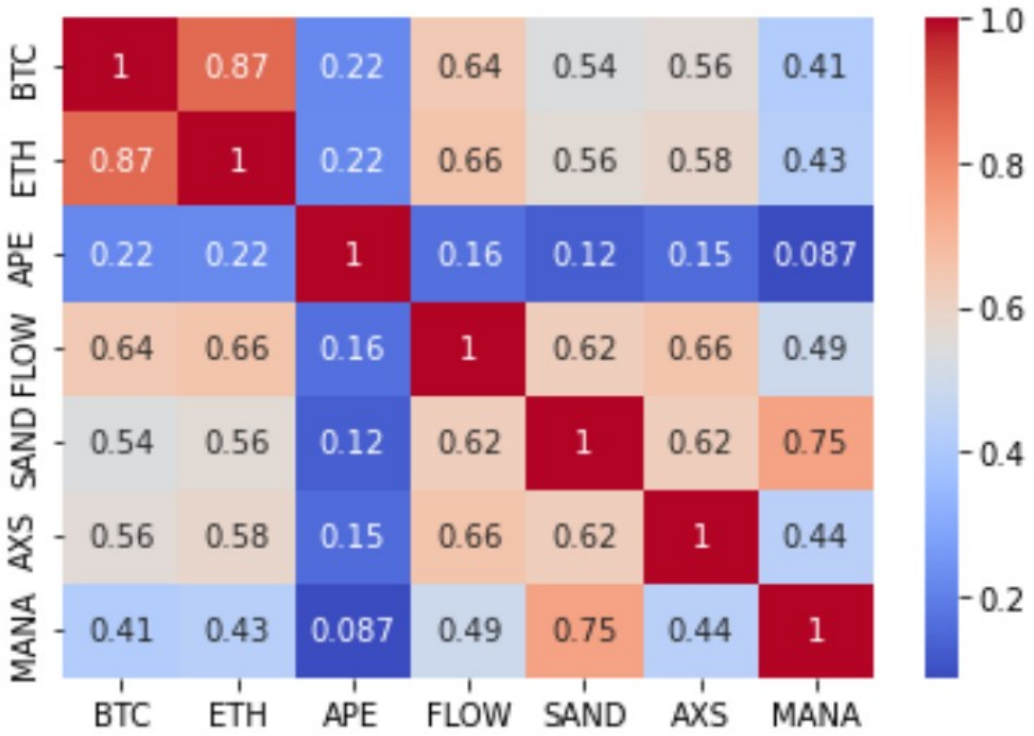


Table 10: Pearson correlation coefficients between cryptoassets

In this later window the coefficients between BTC, ETH and the other cryptocurrencies are considerably higher, both majors displaying a clear linear relationship with the NFT and metaverse assets. APE remains the exception, its coefficients staying low throughout, an indication that its price is little affected by BTC, ETH or the other assets in the study, ApeCoin having established itself as an exclusive, members-only NFT project whose fortunes run somewhat apart from the market.

Discussion

The regression results, taken together, indicate that the linear regression model predicts the Bitcoin price with greater accuracy than either the support vector regression or the neural network. The neural network nonetheless performed respectably, and its accuracy may plausibly be improved by the adjustment of parameters within the model, a matter to which the conclusion returns.

The correlation analysis counsels caution before generalisation. The relationship between Bitcoin and other assets shifts over time and resists confident prediction; that between Bitcoin and other cryptocurrencies varies likewise, under the influence of many factors, with Bitcoin occupying the dominant position. Some cryptocurrencies resemble Bitcoin closely; others serve specific purposes and stand at a further remove from the general market. The analysis of BTC and ETH prices reveals a strong linear relationship between the two, and while certain emerging cryptocurrencies display a weaker linear connection with BTC, the majority of the cryptoassets examined remain strongly tied to its price.

Two qualifications deserve emphasis. The first is the familiar one: correlation is not causation. That two variables move together establishes a relationship between them; it does not disclose the mechanism, nor license the inference that the one drives the other. The second concerns stability: the strength of the linear relationship between BTC and ETH prices varies through time and answers to other influences, market demand and general market conditions among them, so that the relationship requires regular reassessment, and conclusions drawn from any single window must be held provisionally. While the prices of BTC, ETH, the wider cryptoasset market and certain traditional assets may exhibit strong linear relationships, a careful analysis and interpretation of the data remains the condition of any deeper understanding of the forces underlying them (IMF Blog, 2022).

Conclusion

The valuation of a cryptoasset remains a formidable undertaking, for there exists neither an established market convention nor an agreed set of fundamental metrics upon which valuation might rest. Investors and analysts are accordingly driven to combine the income, market and cost approaches in estimating what a cryptoasset is worth.

Regression analysis earns its place within that combination. By modelling the relation between the value of a cryptoasset and independent variables such as market demand and utility, it permits the analyst to assess the influence of those variables upon value and to form predictions of its future course. The present results favour the linear regression model, yet it cannot be asserted with finality that linear regression is the superior instrument for predicting the price of Bitcoin or of any other cryptoasset: such a judgment would require fuller information as to the data, the model parameters and the evaluation metrics employed, and different models will perform differently across datasets according to the complexity of the relation between input and output. Neural networks, in general, possess the capacity to learn complex non-linear relationships and can attain high performance across many tasks; they are, however, sensitive to the choice of hyperparameters, the learning rate and the number of hidden layers among them, and may require careful tuning before that performance is realised. Linear regression, confined as it is to linear relationships, may fail to capture the full complexity of the data; but it is simpler to train, more readily interpreted, and a sound point of departure for more elaborate models. The choice among them must in the end answer to the data and the evaluation criteria at hand, weighed with due regard to the trade-offs between complexity, performance and interpretability.

As to the relation between cryptoassets and the traditional markets, the Pearson correlation coefficient affords the more reliable indication, though its use is complicated by the scarcity of data for most cryptocurrencies. It bears remark that cryptoasset prices have moved under the influence of traditional assets since the onset of the COVID-19 pandemic in March 2020, and under that of institutional investment in blockchain projects.

A combination of the approaches surveyed in this report offers the most comprehensive view of a cryptoasset’s value and the soundest footing for investment decisions. The volatility of these

assets nonetheless enjoins humility: valuations fluctuate substantially and quickly, and any method employed must be applied with caution and revisited as market conditions change.

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Appendix A: Python Code for Regression Analysis and Correlation

```python
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
Created on Tue Dec 10 12:12:55 2022

@author: yas
"""
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

from sklearn.svm import SVR
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense, LSTM, Dropout
from keras.layers import Input, Dense
sns.set()


#DATA VISUALIZATION
csv_path = "/Users/yas/Documents/Dissertation Research/Regression Analysis/BTC/BTC/BTC-USD.csv"
df=pd.read_csv(csv_path, parse_dates=['Date'])
df=df.sort_values('Date')
print(df)

ax=df.plot(x='Date', y='Close')
ax.set_title("Bitcoin Price Over Time")
ax.set_xlabel("Date")
ax.set_ylabel("Price in US Dollar")
plt.show()

shifted_rows = 30
df['Prediction']=df[['Close']].shift(-shifted_rows)
print(df['Prediction'])

#Create Training data to train data for X
training_data = np.array(df[['Close']][:-shifted_rows])
print(training_data)

#Create Training data to train data for Y
y=(df['Prediction'].values)[:-shifted_rows]
print(y)

x_Train, x_Test, y_Train, y_Test=train_test_split(training_data, y, test_size=20)
```

```python
#Linear Regression Model
linReg=LinearRegression()
linReg.fit(x_Train,y_Train)
linReg_confidence=linReg.score(x_Test,y_Test) #Get the score and vale between 0 and 1
print(linReg_confidence)

#Testing the model
future_values=np.array(df[['Close']][-shifted_rows:])
linReg_prediction=linReg.predict(future_values) #Predict the future

#plot the results
plt.plot(future_values, 'red', label='Real Price')
plt.plot(linReg_prediction, 'blue', label='Predicted Price')
plt.xlabel('Time frame')
plt.ylabel('Price in US Dollar')
plt.title('BTC Price Prediction (Linear Regression Model)')
plt.legend()
plt.show()
```

```
#SVR Model
# Define two different
SVR_Model1_RBF=SVR(kernel='rbf', C=1e2, gamma=0.01)
SVR_Model2_POL=SVR(kernel='poly',C=1e3, degree=2)
y_RBF=SVR_Model1_RBF.fit(x_Train, y_Train)
y_POL=SVR_Model2_POL.fit(x_Train, y_Train)

#predicting future values
future_values=np.array(df[['Close']][-shifted_rows:])
y_RBF_prediction=SVR_Model1_RBF.predict(future_values)
y_POL_prediction=SVR_Model2_POL.predict(future_values)

#plot the results
plt.plot(future_values, 'red', label='Original')
plt.plot(y_RBF_prediction,'green', label='SVR_RBF')
plt.plot(y_POL_prediction,'blue', label='SVR_POL')
plt.xlabel('Time frame')
plt.ylabel('Price in US Dollar')
plt.title('BTC Price Prediction (Support Vector Model)')
plt.legend()
plt.show()
```

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
Created on Tue Dec 13 03:01:21 2022

@author: yas
"""

import pandas_datareader as web
import matplotlib.pyplot as plt
import mplfinance as mpf
import seaborn as sns
import datetime as dt

currency = "USD"
metric = "Close"

start = dt.datetime(2020,1,1)
end = dt.datetime.now()

crypto = ['BTH','ETH','TSLA', 'AAPL', 'AMZN', 'GOOGL','META']
colnames = []

first = True
for ticker in crypto:
    data = web.DataReader(f'{ticker}-{currency}', "yahoo", start, end)
    if first:
        combined = data[[metric]].copy()
        colnames.append(ticker)
        combined.columns=colnames
        first=False
    else:
            combined = combined.join(data[metric])
            colnames.append(ticker)
            combined.columns=colnames

combined = combined.pct_change().corr(method="pearson")

sns.heatmap(combined, annot=True, cmap="coolwarm")
plt.show()
```

```
6   @author: yas
7   """
8
9   import pandas_datareader as web
10  import matplotlib.pyplot as plt
11  import mplfinance as mpf
12  import seaborn as sns
13  import datetime as dt
14
15  currency = "USD"
16  metric = "Close"
17
18  start = dt.datetime(2018,1,1)
19  end = dt.datetime(2022,11,30)
20
21  crypto = ['BTC','ETH','XRP','DOT','SOL','LINK','ADA']
22
23  colnames =  []
24
25  first = True
26  for ticker in crypto:
27      data = web.DataReader(f'{ticker}-{currency}', "yahoo", start, end)
28      if first:
29          combined = data[[metric]].copy()
30          colnames.append(ticker)
31          combined.columns=colnames
32          first=False
33      else:
34              combined = combined.join(data[metric])
35              colnames.append(ticker)
36              combined.columns=colnames
37
38  plt.yscale('log')
39
40  for ticker in crypto:
41      plt.plot(combined[ticker], label = ticker)
42
43  plt.legend(loc="upper right")
44
45  plt.show()
```

```python
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
Created on Sun Dec 11 08:14:55 2022

@author: yas
"""

import pandas_datareader as web
import matplotlib.pyplot as plt
import mplfinance as mpf
import seaborn as sns
import datetime as dt

currency = "USD"
metric = "Close"

start = dt.datetime(2020,3,1)
end = dt.datetime(2021,6,1)

crypto = ['BTC','ETH','APE','FLOW','SAND','AXS','MANA']
colnames = []

first = True
for ticker in crypto:
    data = web.DataReader(f'{ticker}-{currency}', "yahoo", start, end)
    if first:
        combined = data[[metric]].copy()
        colnames.append(ticker)
        combined.columns=colnames
        first=False
    else:
            combined = combined.join(data[metric])
            colnames.append(ticker)
            combined.columns=colnames

combined = combined.pct_change().corr(method="pearson")

sns.heatmap(combined, annot=True, cmap="coolwarm")
plt.show()
```

```python
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
Created on Sun Dec 11 07:12:03 2022

@author: yas
"""

import pandas_datareader as web
import matplotlib.pyplot as plt
import mplfinance as mpf
import seaborn as sns
import datetime as dt

currency = "USD"
metric = "Close"

start = dt.datetime(2020,3,1)
end = dt.datetime(2021,6,1)

crypto = ['BTC','ETH','APE','FLOW','SAND','AXS','MANA']
colnames =  []

first = True
for ticker in crypto:
    data = web.DataReader(f'{ticker}-{currency}', "yahoo", start, end)
    if first:
        combined = data[[metric]].copy()
        colnames.append(ticker)
        combined.columns=colnames
        first=False
    else:
            combined = combined.join(data[metric])
            colnames.append(ticker)
            combined.columns=colnames

plt.yscale('log')

for ticker in crypto:
    plt.plot(combined[ticker], label = ticker)

plt.legend(loc="upper right")

plt.show()

```

Appendix B: Fundamental Valuation Methods

Cryptoassets, cryptocurrencies among them, are digital assets that employ cryptography for security and are decentralised in nature, and their valuation is a task of corresponding complexity, for they resemble neither the stocks nor the bonds for which well-established valuation methods exist. Several methods are available, but none is definitive: different methods may produce widely divergent valuations for the same asset, the value of cryptocurrency being highly speculative, extremely volatile, heavily dependent upon market forces and, in consequence, difficult to predict. The principal methods are set out below.

1. Market capitalisation. The most commonly employed measure, market capitalisation is calculated by multiplying the total number of units of a cryptoasset by its current market price. It indicates the size of an asset, though it takes no account of the underlying technology or fundamentals. The concept derives from equity markets, where the market capitalisation of a company is the product of its outstanding shares and the prevailing share price: a company with one million shares trading at $10 has a market capitalisation of $10 million, just as a cryptocurrency with a total supply of ten million tokens trading at $1 apiece is capitalised at $10 million. Formally:

$$\text{Market Cap} = \text{Current Price} \times \text{Circulating Supply}$$

The measure is a rough gauge of size and value, useful for comparison across companies or assets, but imperfect: it reflects neither future earnings potential nor the underlying value of the assets held, and an asset of large capitalisation may possess a lower intrinsic value than one of smaller.

Two variables chiefly govern a cryptocurrency's market capitalisation: supply and price. The supply in global circulation bears upon market price and hence upon capitalisation. Bitcoin affords the canonical example, its protocol, written into the code by its pseudonymous creator Satoshi Nakamoto, capping the total stock at 21 million coins, beyond which no further bitcoins will be mined. The stock of Bitcoin is thus a known and fixed quantity, in contrast even with gold, of whose terrestrial abundance and minable reserves no exact account can be given; and it is this engineered scarcity that underpins its value. Price contributes symmetrically: a

cryptocurrency with few coins in circulation but a high unit price may nonetheless command a large capitalisation (100,000 coins at $10,000 apiece yield a market capitalisation of $1 billion), while a vast circulating supply of coins each worth little produces a modest one, as with a currency of 500,000,000 coins at $0.00005, capitalised at a mere $25,000. The meme coins Dogecoin and Shiba Inu carry very low prices upon massive supplies, where Bitcoin's higher unit price reflects its scarcity; yet since cryptoassets are exceptionally divisible (one may purchase a ten-thousandth part of a bitcoin), the price per coin matters rather less than is commonly supposed.

Capitalisation carries practical consequences. Assets of larger capitalisation tend to attract more investors, being accounted the more appealing, and it is upon capitalisation rather than price that cryptocurrency rankings properly rest, price alone being an insufficient index of market position. Movements in capitalisation may also disclose patterns and market trends: fluctuation in a cryptocurrency associated with non-fungible tokens, for instance, is indicative of the asset attracting or shedding investors.

2. Comparable market analysis. This method values an asset by comparison with similar assets in the market, the essential task being to identify assets alike in technology, use case and other material respects, and to employ their market prices as reference points. The technique is of general application: the real-estate investor values a property by reference to the prices of comparable properties in the same district, and the equity investor a stock by reference to comparable stocks within the industry. Applied to a cryptocurrency, the comparison proceeds upon such factors as total supply, market capitalisation, and adoption and usage across user groups. The method's limitations are those of its inputs: the availability and quality of comparable data govern the accuracy of the valuation, and the selection of comparables can colour the result, for which reasons it is best employed in concert with other methods.

3. Intrinsic value. This method attempts to determine the fundamental value of a cryptoasset from its underlying technology, use case and cognate factors; this is no straightforward task, the intrinsic value of cryptoassets being seldom easy to discern. Intrinsic value denotes the perceived or calculated worth of an asset independent of its market price, resting upon underlying characteristics such as cash flow, earnings potential and growth prospects: an investor may judge a stock intrinsically worth $100 a share upon its earnings prospects while the market prices it at

$50, and conclude that it is undervalued with room to grow. The concept is, however, irreducibly subjective, varying with the investor's perspective and assumptions, and resistant to precise determination, resting as it does upon estimates and projections. For a cryptoasset, intrinsic value is a theoretical quantity reflecting fundamental characteristics, among them usefulness, scarcity and utility, upon the principle that value should rest upon inherent qualities rather than upon market price or production cost alone. No single agreed method of calculation exists, and analysts commonly blend approaches, market capitalisation, cost of production and technical analysis among them, in forming their estimate; the result should inform, but not exhaust, the investment decision.

4. Net asset value (NAV). This method values a cryptoasset by dividing the total value of the assets it holds, whether blockchain technology, intellectual property or partnerships, by the number of units in circulation. The concept is borrowed from the investment fund, whose NAV is computed by subtracting liabilities from assets and dividing by the shares outstanding: a fund holding $10 million of assets against $2 million of liabilities, with one million shares on issue, has a NAV of $8 per share, each share representing ownership of $8 of fund assets. The measure is a serviceable rough gauge of performance and a basis for comparison across funds, though an imperfect one, blind as it is to future earnings potential and to the underlying value of the assets held. Cryptoassets, generating neither cash flows nor earnings in the conventional manner, do not admit the standard NAV computation; a modified version is nonetheless available, in which the value of the asset's underlying holdings, the blockchain network or the technology beneath it, is estimated, liabilities such as outstanding legal claims or unpaid expenses are deducted, and the balance is divided by total supply to yield a NAV per token. The approach labours under evident difficulties, for the value of a cryptoasset's underlying holdings resists accurate estimation and the asset's price is volatile and at the mercy of the market; it should therefore constitute one element of a larger valuation exercise rather than its whole.

5. Earnings-based valuation. This method values a cryptoasset upon its projected future earnings and revenue streams. In its general form, earnings-based valuation estimates an asset's future cash flows and discounts them to present value at a rate reflecting the time value of money and the risk attaching to the asset, the procedure by which a firm might appraise a new product from its forecast cash flows. The method's accuracy depends upon the reliability of the forecasts

and the assumptions embedded in the discount rate, and it is accordingly used in conjunction with other methods. Since cryptoassets do not generate cash flows or earnings in the traditional manner, the standard procedure cannot be applied to them without modification; a modified version, estimating the asset's prospective flows and discounting them at a suitably risk-adjusted rate, yields an estimate of potential value grounded in expected earnings. The difficulties of forecasting a cryptoasset's flows, and the volatility of its value, once again counsel that the method inform rather than determine the final valuation.

6. Discounted cash flow (DCF) analysis. This method values a cryptoasset upon its expected future cash flows discounted to present value at a chosen rate. DCF analysis in its classical form projects the future cash flows of an asset and discounts them at a rate reflecting the time value of money and the level of risk; its accuracy is hostage to the reliability of the projections and the assumptions of the discount rate, and it is best corroborated by other methods. As with the earnings-based approach, the absence of conventional cash flows renders the standard technique inapplicable to cryptoassets in unmodified form; the modified variant estimates prospective flows attributable to the asset and discounts them appropriately, producing an estimate of potential value that shares both the logic and the limitations of its parent method, namely the difficulty of forecasting and the volatility of the underlying, and that should therefore be weighed as one factor among several.

7. Technical analysis. This method analyses historical price data and market trends in search of buying and selling opportunities. The technical analyst studies the historical price and trading data of a cryptoasset to identify patterns and trends from which predictions of future price movements may be drawn, employing an apparatus of chart patterns and indicators to that end. It stands apart from the fundamental methods surveyed above in resting not upon what the asset is, but upon how its market has behaved.